Chapter 1

Vector Analysis

Problem 1.1

Show that the dot product and cross product is distributive

Solution:

Let the three vectors be

$$\boldsymbol{v}_1 = [a, b, c]$$
$$\boldsymbol{v}_2 = [d, e, f]$$
$$\boldsymbol{v}_3 = [g, h, i]$$

dot product:

Show

$$\boldsymbol{v}_1 \cdot (\boldsymbol{v}_2 + \boldsymbol{v}_3) = \boldsymbol{v_1} \cdot \boldsymbol{v}_2 + \boldsymbol{v_1} \cdot \boldsymbol{v}_3$$

Using

$$\boldsymbol{v}_2 + \boldsymbol{v}_3 = [d + g, e + h, f + i]$$
$$\boldsymbol{v}_1 \cdot \boldsymbol{v}_2 = ad + be + cf$$
$$\boldsymbol{v}_1 \cdot \boldsymbol{v}_3 = ag + bh + ci$$

LHS:

$$\boldsymbol{v}_1 \cdot (\boldsymbol{v}_2 + \boldsymbol{v}_3) = [a, b, c] \cdot [d + g, e + h, f + i] =$$
$$= a(d + g) + b(e + h) + c(f + i)$$

RHS:

$$\boldsymbol{v}_1 \cdot \boldsymbol{v}_2 + \boldsymbol{v}_1 + \boldsymbol{v}_3 = ad + be + cf + ag + bh + ci =$$
$$= a(d + g) + b(e + h) + c(f + i)$$

cross product: Show

$$\boldsymbol{v}_1 \times (\boldsymbol{v}_2 + \boldsymbol{v}_3) = (\boldsymbol{v}_1 \times \boldsymbol{v}_2) + (\boldsymbol{v}_1 \times \boldsymbol{v}_3)$$

Using

$$\boldsymbol{v}_2 + \boldsymbol{v}_3 = [d+g, e+h, f+i]$$
$$\boldsymbol{v}_1 \times \boldsymbol{v}_2 = bf - ce + cd - af + ae - bd$$
$$\boldsymbol{v}_1 \times \boldsymbol{v}_3 = bi - ch + cg - ai + ah - bg$$

which gives

$$(\boldsymbol{v}_1 \times \boldsymbol{v}_2) + (\boldsymbol{v}_1 \times \boldsymbol{v}_3) = a(e+h-f-i) + b(f+i-d-g) + c(d+g-e-h)$$

and LHS:

$$\begin{aligned} \boldsymbol{v}_1 \times (\boldsymbol{v}_2 + \boldsymbol{v}_3) &= b(f+i) - c(e+h) + c(d+g) - a(f+i) + a(e+h) - b(d+g) = \\ &= bf + bi - ce - ch + cd + cg - af - ai + ae + ah - bd - bg = \\ &= a(e+h-f-i) + b(f+i-d-g) + c(d+g-e-h) \end{aligned}$$

□

Problem 1.2

Is the cross product associative?

$$(\boldsymbol{A} \times \boldsymbol{B}) \times \boldsymbol{C} \stackrel{?}{=} \boldsymbol{A} \times (\boldsymbol{B} \times \boldsymbol{C})$$

Solution:

No, for example set

$$\boldsymbol{A} = \hat{\mathbf{x}}$$
$$\boldsymbol{B} = \hat{\mathbf{x}}$$
$$\boldsymbol{C} = \hat{\mathbf{z}}$$

Then, the left hand side above becomes

$$(\boldsymbol{A} \times \boldsymbol{B}) \times \boldsymbol{C} = (\hat{\mathbf{x}} \times \hat{\mathbf{x}}) \times \hat{\mathbf{z}} = 0$$

while the right hand side is

$$\boldsymbol{A} \times (\boldsymbol{B} \times \boldsymbol{C}) = \hat{\mathbf{x}} \times (\hat{\mathbf{x}} \times \hat{\mathbf{z}}) = \hat{\mathbf{x}} \times -\hat{\mathbf{y}} = -\hat{\mathbf{z}}$$

□

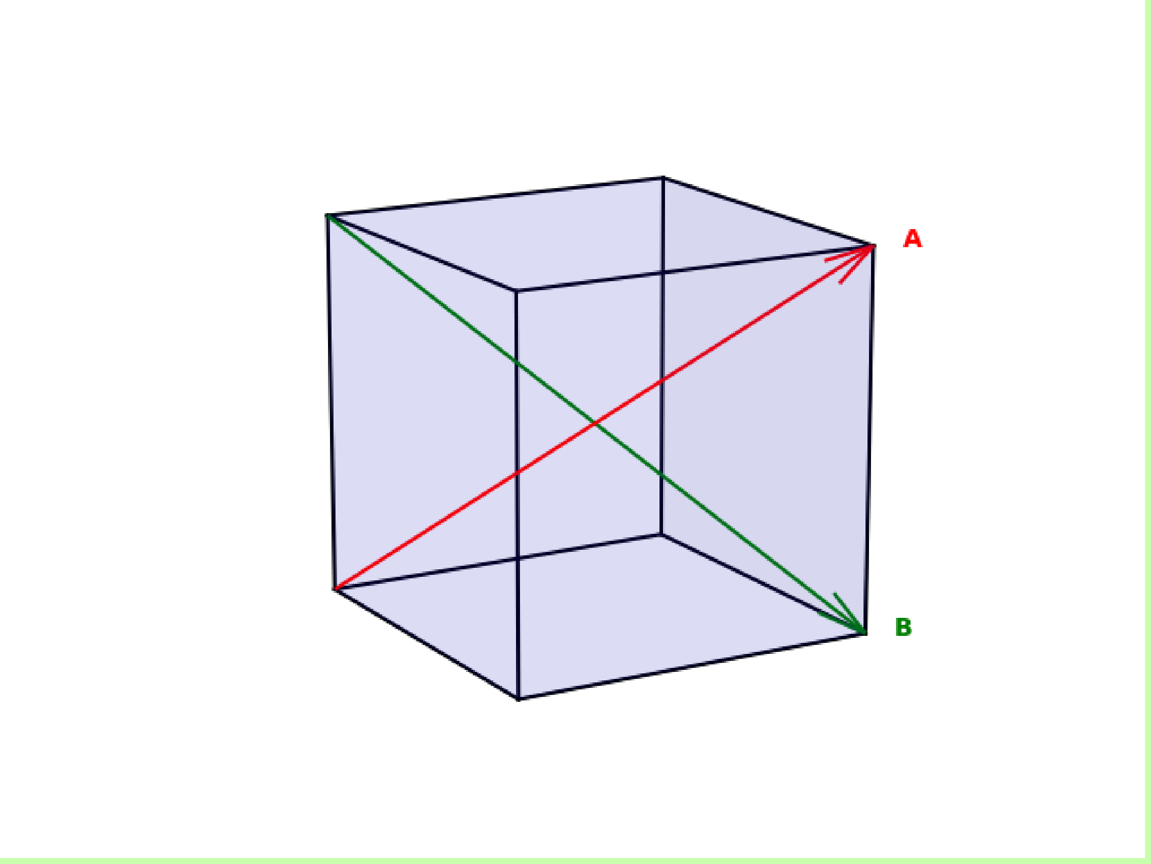
Problem 1.3

Find the angle between the body diagonals of a cube.

Solution:

$$\frac{\partial^{k+4} x}{\partial y^2 \partial z^k \partial p^2}$$
$$\boldsymbol{\nabla} \times B$$
$$\left(\frac{\partial B_z}{\partial y} - \frac{\partial B_y}{\partial z}\right)\hat{\mathbf{x}} + \left(\frac{\partial B_x}{\partial z} - \frac{\partial B_z}{\partial x}\right)\hat{\mathbf{y}} + \left(\frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y}\right)\hat{\mathbf{z}}$$

The body diagonals $\boldsymbol{A} = [1, 1, 1]$ and $\boldsymbol{B} = [1, 1, -1]$



$|\boldsymbol{A}| = |\boldsymbol{B}| = \sqrt{3}$

$\boldsymbol{A} \cdot \boldsymbol{B} = 1 + 1 - 1 = 1$

$\boldsymbol{A} \cdot \boldsymbol{B} = |\boldsymbol{A}|\ |\boldsymbol{B}| \cos\theta = 3\cos\theta = 1 \rightarrow \theta = \arccos\left(\frac{1}{3}\right) \approx 70.53° \square$

Problem 1.4

Use the cross product to find the components of the unit vector $\hat{\mathbf{n}}$ perpendicular to the shaded plane in Fig.

Solution:

Two of the plane vectors are $\boldsymbol{v}_1 = [-1, 2, 0]$ and $\boldsymbol{v}_2 = [-1, 0, 3]$. $\boldsymbol{v}_1 \times \boldsymbol{v}_2 = n = [6, 3, 2]$ and $|\boldsymbol{n}| = 7$ gives $\hat{\mathbf{n}} = \frac{1}{7}[6, 3, 2]$ $\square$

Problem 1.5

Prove the **BAC-CAB** rule,

$$\boldsymbol{A} \times (\boldsymbol{B} \times \boldsymbol{C}) = \boldsymbol{B}(\boldsymbol{A} \cdot \boldsymbol{C}) - \boldsymbol{C}(\boldsymbol{A} \cdot \boldsymbol{B})$$

by writing out both sides in component form.

Solution:

Writing out the vectors in component form

$$\boldsymbol{A} = A_x\hat{\mathbf{x}} + A_y\hat{\mathbf{y}} + A_z\hat{\mathbf{z}}$$
$$\boldsymbol{B} = B_x\hat{\mathbf{x}} + B_y\hat{\mathbf{y}} + B_z\hat{\mathbf{z}}$$
$$\boldsymbol{C} = C_x\hat{\mathbf{x}} + C_y\hat{\mathbf{y}} + C_z\hat{\mathbf{z}}$$

starting with lhs

$$\boldsymbol{B} \times \boldsymbol{C} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ B_x & B_y & B_z \\ C_x & C_y & C_z \end{vmatrix} = \left(B_yC_z - B_zC_y\right)\hat{\mathbf{x}} + (B_zC_x - B_xC_z)\hat{\mathbf{y}} + \left(B_xC_y - B_yC_x\right)\hat{\mathbf{z}}$$

rewriting with a new matrix $\boldsymbol{D}$ where

$$D_x = B_yC_z - B_zC_y$$
$$D_y = B_zC_x - B_xC_z$$
$$D_z = B_xC_y - B_yC_x$$

gives

$$\boldsymbol{D} = \boldsymbol{B} \times \boldsymbol{C} = D_x\hat{\mathbf{x}} + D_y\hat{\mathbf{y}} + D_z\hat{\mathbf{z}}$$

Now

$$\boldsymbol{A} \times (\boldsymbol{B} \times \boldsymbol{C}) = \boldsymbol{A} \times \boldsymbol{D} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ A_x & A_y & A_z \\ D_x & D_y & D_z \end{vmatrix} =$$
$$= \left(A_yD_z - A_zD_y\right)\hat{\mathbf{x}} + (A_zD_x - A_xD_z)\hat{\mathbf{y}} + \left(A_xD_y - A_yD_x\right)\hat{\mathbf{z}}$$

RHS:

$$\boldsymbol{A}\cdot\boldsymbol{C} = A_xC_x + A_yC_y + A_zC_z = P$$
$$\boldsymbol{A}\cdot\boldsymbol{B} = A_xB_x + A_yB_y + A_zB_z = Q$$
$$\boldsymbol{B}(\boldsymbol{A}\cdot\boldsymbol{C}) = B_xP\hat{\mathbf{x}} + B_yP\hat{\mathbf{y}} + B_zP\hat{\mathbf{z}}$$
$$\boldsymbol{C}(\boldsymbol{A}\cdot\boldsymbol{B}) = C_xQ\hat{\mathbf{x}} + C_yQ\hat{\mathbf{y}} + C_zQ\hat{\mathbf{z}}$$
$$\boldsymbol{B}(\boldsymbol{A}\cdot\boldsymbol{C}) - \boldsymbol{C}(\boldsymbol{A}\cdot\boldsymbol{B}) = (B_xP - C_xQ)\hat{\mathbf{x}} + \left(B_yP - C_yQ\right)\hat{\mathbf{y}} + (B_zP - C_zQ)\hat{\mathbf{z}}$$

Now final step is checking if

$$\mathbf{1}\text{: } A_yD_z - A_zD_y = B_xP - C_xQ$$
$$\mathbf{2}\text{: } A_zD_x - A_xD_z = B_yP - C_yQ$$
$$\mathbf{3}\text{: } A_xD_y - A_yD_x = B_zP - C_zQ$$

For **1**:

$$\begin{aligned}\text{LHS: } &A_y\left(B_xC_y - B_yC_x\right) - A_z(B_zC_x - B_xC_z)\\ &= A_yB_xC_y - A_yB_yC_x - A_zB_zC_x + A_zB_xC_z\\ &= B_x\left(A_yC_y + A_zC_z\right) - C_x\left(A_yB_y + A_zB_z\right)\end{aligned}$$

$$\begin{aligned}\text{RHS: } &B_x\left(A_xC_x + A_yC_y + A_zC_z\right) - C_x\left(A_xB_x + A_yB_y + A_zB_z\right)\\ &= B_xA_xC_x + B_x\left(A_yC_y + A_zC_z\right) - C_xA_xB_x - C_x\left(A_yB_y + A_zB_z\right)\\ &= B_x\left(A_yC_y + A_zC_z\right) - C_x\left(A_yB_y + A_zB_z\right) = \text{LHS } \checkmark\end{aligned}$$

For **2**:

$$\begin{aligned}\text{LHS: } &A_z\left(B_yC_z - B_zC_y\right) - A_x\left(B_xC_y - B_yC_x\right)\\ &= A_zB_yC_z - A_zB_zC_y - A_xB_xC_y + A_xB_yC_x\\ &= B_y(A_zC_z + A_xC_x) - C_y(A_zB_z + A_xB_x)\end{aligned}$$

$$\begin{aligned}\text{RHS: } &B_y\left(A_xC_x + A_yC_y + A_zC_z\right) - C_y\left(A_xB_x + A_yB_y + A_zB_z\right)\\ &= B_yA_yC_y + B_y(A_xC_x + A_zC_z) - C_yA_yB_y - C_y(A_xB_x + A_zB_z)\\ &= B_y(A_xC_x + A_zC_z) - C_y(A_xB_x + A_zB_z) = \text{LHS } \checkmark\end{aligned}$$

For **3**:

$$\begin{aligned}\text{LHS: } &A_x(B_zC_x - B_xC_z) - A_y\left(B_yC_z - B_zC_y\right)\\ &= A_xB_zC_x - A_xB_xC_z - A_yB_yC_z + A_yB_zC_y\\ &= B_z\left(A_xC_x + A_yC_y\right) - C_z\left(A_xB_x + A_yB_y\right)\end{aligned}$$

$$\begin{aligned}\text{RHS: } &B_z\left(A_xC_x + A_yC_y + A_zC_z\right) - C_z\left(A_xB_x + A_yB_y + A_zB_z\right)\\ &= B_zA_zC_z + B_z\left(A_xC_x + A_yC_y\right) - C_zA_zB_z - C_z\left(A_xB_x + A_yB_y\right)\\ &= B_z\left(A_xC_x + A_yC_y\right) - C_z\left(A_xB_x + A_yB_y\right) = \text{LHS } \checkmark\end{aligned}$$

Since all three components are equal, the BAC-CAB rule is proven. □

Problem 1.6

Prove that

$$[\boldsymbol{A} \times (\boldsymbol{B} \times \boldsymbol{C})] + [\boldsymbol{B} \times (\boldsymbol{C} \times \boldsymbol{A})] + [\boldsymbol{C} \times (\boldsymbol{A} \times \boldsymbol{B})] = 0$$

Under what conditions does $\boldsymbol{A} \times (\boldsymbol{B} \times \boldsymbol{C}) = (\boldsymbol{A} \times \boldsymbol{B}) \times \boldsymbol{C}$?

Solution:

Using the BAC-CAB rule, rewrite the triple cross product as

$$\begin{aligned}
(1)\ \boldsymbol{A} \times (\boldsymbol{B} \times \boldsymbol{C}) &= \boldsymbol{B}(\boldsymbol{A} \cdot \boldsymbol{C}) - \boldsymbol{C}(\boldsymbol{A} \cdot \boldsymbol{B}) \\
(2)\ \boldsymbol{B} \times (\boldsymbol{C} \times \boldsymbol{A}) &= \boldsymbol{C}(\boldsymbol{B} \cdot \boldsymbol{A}) - \boldsymbol{A}(\boldsymbol{B} \cdot \boldsymbol{C}) \\
(3)\ \boldsymbol{C} \times (\boldsymbol{A} \times \boldsymbol{B}) &= \boldsymbol{A}(\boldsymbol{C} \cdot \boldsymbol{B}) - \boldsymbol{B}(\boldsymbol{C} \cdot \boldsymbol{A})
\end{aligned}$$

Using the commutative property of the dot product, $\boldsymbol{v}_1 \cdot \boldsymbol{v}_2 = \boldsymbol{v}_2 \cdot \boldsymbol{v}_1$, it is seen that:

- The $\boldsymbol{B}$-part in (1) cancels out the $\boldsymbol{B}$-part in (3)
- The $\boldsymbol{C}$-part in (1) cancels out the $\boldsymbol{C}$-part in (2)
- The $\boldsymbol{A}$-part in (2) cancels out the $\boldsymbol{A}$-part in (3)

The conditions when $\boldsymbol{A} \times (\boldsymbol{B} \times \boldsymbol{C}) = (\boldsymbol{A} \times \boldsymbol{B}) \times \boldsymbol{C}$ is translated with the BAC-CAB rule into $\boldsymbol{B}(\boldsymbol{A} \cdot \boldsymbol{C}) - \boldsymbol{C}(\boldsymbol{A} \cdot \boldsymbol{B}) = -\boldsymbol{A}(\boldsymbol{C} \cdot \boldsymbol{B}) + \boldsymbol{B}(\boldsymbol{C} \cdot \boldsymbol{A})$ which gives that either $\boldsymbol{A} = \boldsymbol{C}$ or $\boldsymbol{B} \perp (\boldsymbol{A}$ and $\boldsymbol{C})$ □

Problem 1.7

Find the separation vector $\mathscr{r} rs$ from the source point (2, 8, 7) to the field point (4, 6, 8). Determine its magnitude $(\mathscr{r})$, and construct the unit vector $\hat{\boldsymbol{\mathscr{r}}}$.

Solution:

The separation vector $\boldsymbol{\mathscr{r}}$ is the difference between the field point and the source point.

$$\boldsymbol{\mathscr{r}} = (4, 6, 8) - (2, 8, 7) = (2, -2, 1)$$

The magnitude is

$$|\boldsymbol{\mathscr{r}}| = \mathscr{r} = \sqrt{2^2 + (-2)^2 + 1^2} = 3$$

and the unit vector is

$$\frac{\boldsymbol{\mathscr{r}}}{\mathscr{r}} = \hat{\boldsymbol{\mathscr{r}}} = \frac{1}{3}(2, -2, 1)$$

□

Problem 1.8

$$\begin{pmatrix} \overline{A}_y \\ \overline{A}_z \end{pmatrix} = \begin{pmatrix} \cos\phi & \sin\phi \\ -\sin\phi & \cos\phi \end{pmatrix} \begin{pmatrix} A_y \\ A_z \end{pmatrix} \tag{1}$$

$$\begin{pmatrix} \overline{A}_x \\ \overline{A}_y \\ \overline{A}_z \end{pmatrix} = \begin{pmatrix} R_{xx} & R_{xy} & R_{xz} \\ R_{yx} & R_{yy} & R_{yz} \\ R_{zx} & R_{zy} & R_{xx} \end{pmatrix} \begin{pmatrix} A_x \\ A_y \\ A_z \end{pmatrix} \tag{2}$$

(a) Prove that the two-dimensional rotation matrix (Eq. 1) dot products. (That is, show that $\overline{A}_y\overline{B}_y + \overline{A}_z\overline{B}_z = A_yB_y + A_zB_z$).

(b) What constraints must the elements (R_{ij}) of the three-dimensional rotation matrix (Eq. 2) satisfy in order to preserve the length of **A** (for all vectors **A**)?

Solution:

(a)

$$\begin{pmatrix}\overline{A}_y\\ \overline{A}_z\end{pmatrix}\cdot\begin{pmatrix}\overline{B}_y\\ \overline{B}_z\end{pmatrix}=\begin{pmatrix}\cos\phi & \sin\phi\\ -\sin\phi & \cos\phi\end{pmatrix}\begin{pmatrix}A_y\\ A_z\end{pmatrix}\cdot\begin{pmatrix}\cos\phi & \sin\phi\\ -\sin\phi & \cos\phi\end{pmatrix}\begin{pmatrix}B_y\\ B_z\end{pmatrix}$$

$$\downarrow$$

$$\overline{A}_y\overline{B}_y+\overline{A}_z\overline{B}_z=\begin{pmatrix}\cos\phi A_y+\sin\phi A_z\\ -\sin\phi A_y+\cos\phi A_z\end{pmatrix}\cdot\begin{pmatrix}\cos\phi B_y+\sin\phi B_z\\ -\sin\phi B_y+\cos\phi B_z\end{pmatrix}=$$

$$=\left(\cos\phi A_y+\sin\phi A_z\right)\left(\cos\phi B_y+\sin\phi B_z\right)+\left(-\sin\phi A_y+\cos\phi A_z\right)\left(-\sin\phi B_y+\cos\phi B_z\right)=$$

$$\begin{gathered}=\cos^2\phi A_yB_y+\cos\phi\sin\phi A_yB_z+\sin\phi\cos\phi A_zB_y+\sin^2 A_zB_z+\\ +\sin^2\phi A_yB_y-\sin\phi\cos\phi A_yB_z-\cos\phi\sin\phi A_zB_y+\cos^2\phi A_zB_z=\end{gathered}$$

$$\begin{gathered}=(\cos^2\phi+\sin^2\phi)A_yB_y+(\cos^2\phi+\sin^2\phi)A_zB_z+\\ +\underbrace{(\cos\phi\sin\phi-\sin\phi\cos\phi)}_{=0}A_yB_z+\underbrace{(\sin\phi\cos\phi-\cos\phi\sin\phi)}_{=0}A_zB_y=\end{gathered}$$

$$\boxed{=A_yB_y+A_zB_z=\overline{A}_y\overline{B}_y+\overline{A}_z\overline{B}_z}$$

(b) We want to preserve the length which means

$$\|\mathbf{R}\cdot\mathbf{A}\|=\|\mathbf{A}\|$$

Squaring both sides gives

$$(\mathbf{RA})^\mathsf{T}(\mathbf{RA})=\mathbf{A}^\mathsf{T}\mathbf{A}$$

$$\downarrow$$

$$\mathbf{A}^\mathsf{T}\mathbf{R}^\mathsf{T}\mathbf{RA}=\mathbf{A}^\mathsf{T}\mathbf{A}$$

which gives

$$\boxed{\mathbf{R}^\mathsf{T}\mathbf{R}=\mathbf{I}=\begin{pmatrix}1&0&0\\0&1&0\\0&0&1\end{pmatrix}}$$

$\square$

Problem 1.9

Problem 1.10

Problem 1.11

Find the gradients of the following functions:

(a) $f(x, y, z) = x^2 + y^3 + z^4$

(b) $f(x, y, z) = x^2y^3z^4$

(c) $f(x, y, z) = e^x \sin(y) \ln(z)$

Solution:

The gradient of a function T is defined as

$$\boldsymbol{\nabla} T = \frac{\partial T}{\partial x}\hat{\mathbf{x}} + \frac{\partial T}{\partial y}\hat{\mathbf{y}} + \frac{\partial T}{\partial z}\hat{\mathbf{z}}$$

This results in

(a) $2x\hat{\mathbf{x}} + 3y^2\hat{\mathbf{y}} + 4z^3\hat{\mathbf{z}}$

(b) $2xy^3z^4\hat{\mathbf{x}} + 3x^2y^2z^4\hat{\mathbf{y}} + 4x^2y^3z^3\hat{\mathbf{z}}$

(c) $e^x\left(\sin(y)\ln(z)\hat{\mathbf{x}} + \cos(y)\ln(z)\hat{\mathbf{y}} + \frac{\sin(y)}{z}\hat{\mathbf{z}}\right)$

□

Problem 1.12

The height of a certain hill (in feet) is given by

$$h(x, y) = 10(2xy - 3x^2 - 4y^2 - 18x + 28y + 12)$$

where y is the distance (in miles) north, x the distance east, of South Hadley.

(a) Where is the top of the hill located?

(b) How high is the hill?

(c) How steep is the slope (in feet per mile) at a point 1 mile north and 1 mile east of South Hadley? In what direction is the slope steepest, at that point?

Solution:

The gradient of $h(x, y)$ is

$$\nabla h = (20y - 60x - 180)\hat{\mathbf{x}} + (20x - 80y + 280)\hat{\mathbf{y}}$$

(a) The location of the top is calculated by setting the gradient to 0. This means

$$20y - 60x - 180 = 0$$
$$20x - 80y + 280 = 0$$

which has the solution $(x, y) = (-2, 3)$ miles.

(b) Putting the values given by (a) into the height function $h(x, y)$, the height of the hill is $h(-2, 3) = 720$ft.

(c) At the point $(x, y) = (1, 1)$, the steepness and direction is obtained by putting this point into the gradient. The slope is then $\sqrt{2} \cdot 220$ ft/mile in the direction $-\hat{\mathbf{x}} + \hat{\mathbf{y}}$, which corresponds to northwest. $\square$

Problem 1.13

Let $\boldsymbol{\mathscr{r}}$ be the separation vector from a fixed point (x', y', z') to the point (x, y, z), and let $\mathscr{r}$ be its length. Show that

$$\text{(a) } \nabla(\mathscr{r}^2) = 2\boldsymbol{\mathscr{r}}$$

(b) $\nabla\left(\frac{1}{\mathscr{r}}\right) = -\hat{\boldsymbol{\mathscr{r}}}/\mathscr{r}^2$

(c) What is the general formula for $\nabla(\mathscr{r}^n)$?

Solution:

$$\boldsymbol{\mathscr{r}} = (x - x', y - y', z - z')$$
$$\mathscr{r} = \sqrt{(x - x')^2 + (y - y')^2 + (z - z')^2}$$

(a)

$$\nabla(\mathscr{r}^2) = \nabla\Big((x - x')^2 + (y - y')^2 + (z - z')^2\Big) =$$
$$= 2(x - x')\hat{\mathbf{x}} + 2(y - y')\hat{\mathbf{y}} + 2(z - z')\hat{\mathbf{z}} = 2\boldsymbol{\mathscr{r}}$$

(b)

$$\nabla\left(\frac{1}{\mathscr{r}}\right) = \nabla\left(\frac{1}{\sqrt{(x-x')^2+(y-y')^2+(z-z')^2}}\right) =$$

$$= -\frac{1}{2}\frac{2(x-x')+2(y-y')+2(z-z')}{\sqrt{(x-x')^2+(y-y')^2+(z-z')^2}^3} = -\frac{\boldsymbol{\mathscr{r}}}{\mathscr{r}^3} = -\frac{\hat{\boldsymbol{\mathscr{r}}}}{\mathscr{r}^2}$$

(c)

$$\nabla(\mathscr{r}^n) = \nabla\left(\left((x-x')^2+(y-y')^2+(z-z')^2\right)^{\frac{n}{2}}\right) =$$

$$= \frac{n}{2}(2(x-x')\hat{\mathbf{x}}+2(y-y')\hat{\mathbf{y}}+2(z-z')\hat{\mathbf{z}})\left[(x-x')^2+(y-y')^2+(z-z')^2\right]^{\frac{n-2}{2}} =$$

$$= n\mathscr{r}^{n-2}\boldsymbol{\mathscr{r}}$$

□

Problem 1.14

Problem 1.15

Calculate the divergence of the following vector functions:

(a) $\boldsymbol{v}_a = x^2\hat{\mathbf{x}} + 3xz^2\hat{\mathbf{y}} - 2xz\hat{\mathbf{z}}$

(b) $\boldsymbol{v}_b = xy\hat{\mathbf{x}} + 2yz\hat{\mathbf{y}} + 3xz\hat{\mathbf{z}}$

(c) $\boldsymbol{v}_c = y^2\hat{\mathbf{x}} + (2xy + z^2)\hat{\mathbf{y}} + 2yz\hat{\mathbf{z}}$

Solution:

The divergence of a vector $\boldsymbol{v}$ is defined as

$$\nabla\cdot\boldsymbol{v} = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}$$

(a) $\nabla\cdot\boldsymbol{v}_a = 2x - 2x = 0$

(b) $\nabla\cdot\boldsymbol{v}_b = y + 2z + 3x$

(c) $\nabla\cdot\boldsymbol{v}_c = 2x + 2y$

□

Problem 1.16

Sketch the vector function

$$\boldsymbol{v} = \frac{\hat{\mathbf{r}}}{r^2},$$

and compute its divergence.

Solution:

The divergence of the r-component in spherical coordinates is

$$\boldsymbol{\nabla} \cdot \boldsymbol{v} = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 v_r)$$

Using this, the divergence is $\boldsymbol{\nabla} \cdot \boldsymbol{v} = 0$. From the sketch, clearly the divergence is not 0 at the origin. The problem arises at the point $\boldsymbol{r} = 0$ where $\boldsymbol{v} \to \infty$. At all other points $\boldsymbol{r} \neq 0$, the divergence is 0.

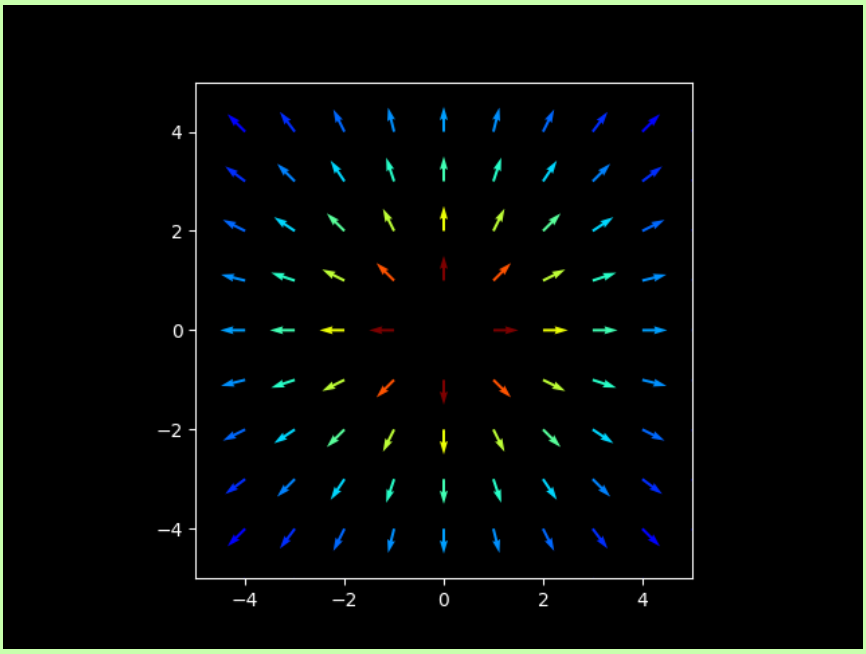


Figure 2: Sketch of the function $\boldsymbol{v} = \frac{\hat{\mathbf{r}}}{r^2}$

□

Problem 1.17

Problem 1.18

Calculate the curls of the vector functions

(a) $\boldsymbol{v}_a = x^2\hat{\mathbf{x}} + 3xz^2\hat{\mathbf{y}} - 2xz\hat{\mathbf{z}}$

(b) $\boldsymbol{v}_b = xy\hat{\mathbf{x}} + 2yz\hat{\mathbf{y}} + 3xz\hat{\mathbf{z}}$

(c) $\boldsymbol{v}_c = y^2\hat{\mathbf{x}} + (2xy + z^2)\hat{\mathbf{y}} + 2yz\hat{\mathbf{z}}$

Solution:

The definition of the curl of a vector function $\boldsymbol{v}$ is

$$\boldsymbol{\nabla} \times \boldsymbol{v} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ v_x & v_y & v_z \end{vmatrix} = \left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z}\right)\hat{\mathbf{x}} + \left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x}\right)\hat{\mathbf{y}} + \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y}\right)\hat{\mathbf{z}}$$

This gives

(a) $\boldsymbol{\nabla} \times \boldsymbol{v}_a = -6xz\hat{\mathbf{x}} + 2z\hat{\mathbf{y}} + 3z^2\hat{\mathbf{z}}$

(b) $\boldsymbol{\nabla} \times \boldsymbol{v}_b = -2y\hat{\mathbf{x}} + -3z\hat{\mathbf{y}} + -x\hat{\mathbf{z}}$

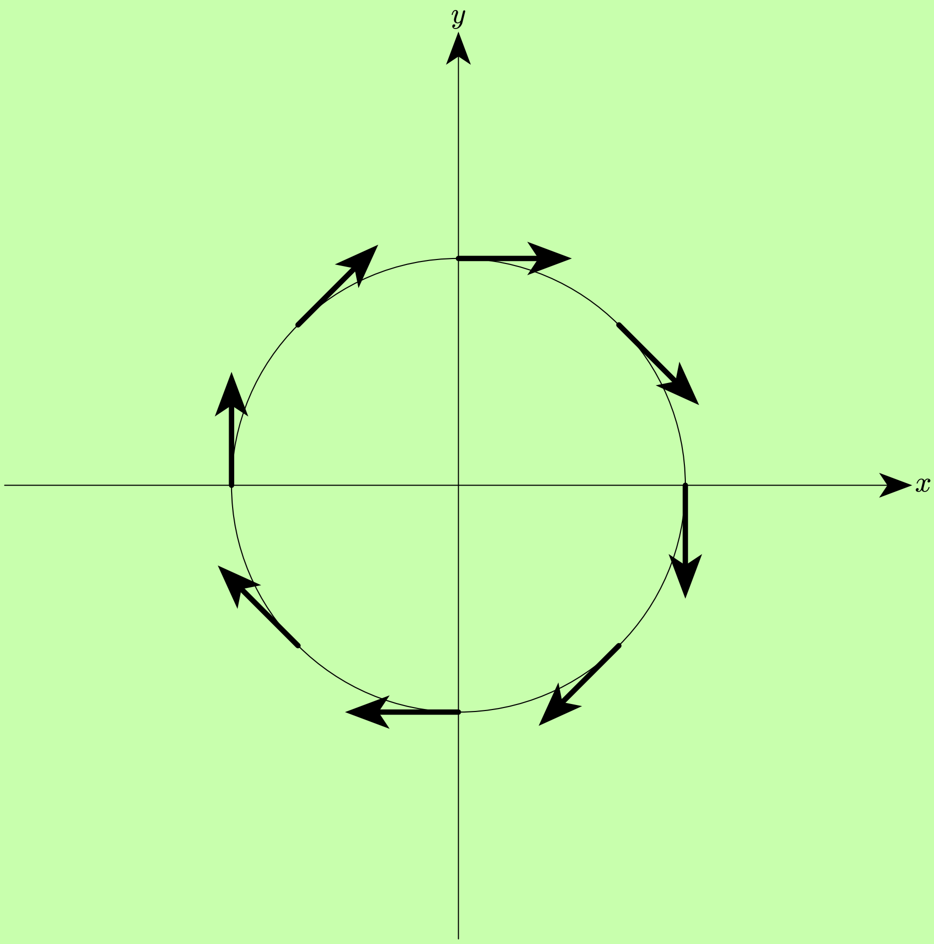
(c) $\boldsymbol{\nabla} \times \boldsymbol{v}_c = 0$

□

Problem 1.19

Draw a circle in the x-y plane. At a few representative points draw the unit vector $\hat{\mathbf{v}}$ tangent to the circle, pointing in the clockwise direction. By comparing adjacent vectors, determine the signs of $\frac{\partial v_x}{\partial y}$ and $\frac{\partial v_y}{\partial x}$. According to The definition of the curl, what is the direction of $\boldsymbol{\nabla} \times \hat{\mathbf{v}}$?

Solution:



Going from one arrow to the next, it is seen that

$$\frac{\partial v_x}{\partial y} = \text{positive}$$

$$\frac{\partial v_y}{\partial x} = \text{negative}$$

For example, consider the arrow at 12 o'clock with $x = 0, y = 1, v_x = 1, v_y = 0$, and the arrow after it at around 2 o'clock with $x = 0.7, y = 0.7, v_x = 0.7, v_y = -0.7$. This confirms the values of $\frac{\partial v_x}{\partial y}$ and $\frac{\partial v_y}{\partial x}$ above.

$\square$

Problem 1.20

Construct a vector function that has a zero divergence and zero curl everywhere.

Solution:

The vector function $\boldsymbol{R}$ should satisfy

$$\begin{aligned}\boldsymbol{\nabla} \cdot \boldsymbol{R} &= 0 \\ \boldsymbol{\nabla} \times \boldsymbol{R} &= 0\end{aligned}$$

Using the definitions of the divergence and curl, the conditions are

$$\boldsymbol{\nabla} \cdot \boldsymbol{R} = \frac{\partial R_x}{\partial x} + \frac{\partial R_y}{\partial y} + \frac{\partial R_z}{\partial z} = 0$$

$$\boldsymbol{\nabla} \times \boldsymbol{R} = \left(\frac{\partial R_z}{\partial y} - \frac{\partial R_y}{\partial z}\right)\hat{\mathbf{x}} + \left(\frac{\partial R_x}{\partial z} - \frac{\partial R_z}{\partial x}\right)\hat{\mathbf{y}} + \left(\frac{\partial R_y}{\partial x} - \frac{\partial R_x}{\partial y}\right)\hat{\mathbf{z}}$$

Want to fulfill:

$$\begin{aligned}\frac{\partial R_z}{\partial y} &= \frac{\partial R_y}{\partial z} \\ \frac{\partial R_x}{\partial z} &= \frac{\partial R_z}{\partial x} \\ \frac{\partial R_y}{\partial x} &= \frac{\partial R_x}{\partial y}\end{aligned}$$

An example is $\boldsymbol{R} = yz\hat{\mathbf{x}} + xz\hat{\mathbf{y}} + xy\hat{\mathbf{z}}$

□

Problem 1.21

Prove the product rules

$$\begin{aligned}&\text{(i)} \quad \boldsymbol{\nabla}(fg) = f\boldsymbol{\nabla}g + g\boldsymbol{\nabla}f \\ &\text{(iv)} \ \boldsymbol{\nabla} \cdot (\boldsymbol{A} \times \boldsymbol{B}) = \boldsymbol{B} \cdot (\boldsymbol{\nabla} \times \boldsymbol{A}) - \boldsymbol{A} \cdot (\boldsymbol{\nabla} \times \boldsymbol{B}) \\ &\text{(v)} \quad \boldsymbol{\nabla} \times (f\boldsymbol{A}) = f(\boldsymbol{\nabla} \times \boldsymbol{A}) - \boldsymbol{A} \times (\boldsymbol{\nabla} f)\end{aligned}$$

Solution:

(i)

$$\nabla(fg) = \frac{\partial}{\partial x}(fg)\hat{\mathbf{x}} + \frac{\partial}{\partial y}(fg)\hat{\mathbf{y}} + \frac{\partial}{\partial z}(fg)\hat{\mathbf{z}}$$

$$\downarrow$$

$$\frac{\partial}{\partial x}(fg)\hat{\mathbf{x}} = \left[g\frac{\partial f}{\partial x} + f\frac{\partial g}{\partial x}\right]\hat{\mathbf{x}}$$

$$\frac{\partial}{\partial y}(fg)\hat{\mathbf{y}} = \left[g\frac{\partial f}{\partial y} + f\frac{\partial g}{\partial y}\right]\hat{\mathbf{y}}$$

$$\frac{\partial}{\partial z}(fg)\hat{\mathbf{z}} = \left[g\frac{\partial f}{\partial z} + f\frac{\partial g}{\partial z}\right]\hat{\mathbf{z}}$$

$$\downarrow$$

$$\boxed{\nabla(fg) = f\nabla g + g\nabla f}$$

(iv) Expand the cross product $\mathbf{A} \times \mathbf{B}$ and the curls $\nabla \times \mathbf{A}, \nabla \times \mathbf{B}$

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix} = (A_yB_z - A_zB_y)\hat{\mathbf{x}} + (A_zB_x - A_xB_z)\hat{\mathbf{y}} + (A_xB_y - A_yB_x)\hat{\mathbf{z}}$$

$$\nabla \times \mathbf{A} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix} = \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right)\hat{\mathbf{x}} + \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right)\hat{\mathbf{y}} + \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right)\hat{\mathbf{z}}$$

$$\nabla \times \mathbf{B} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ B_x & B_y & B_z \end{vmatrix} = \left(\frac{\partial B_z}{\partial y} - \frac{\partial B_y}{\partial z}\right)\hat{\mathbf{x}} + \left(\frac{\partial B_x}{\partial z} - \frac{\partial B_z}{\partial x}\right)\hat{\mathbf{y}} + \left(\frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y}\right)\hat{\mathbf{z}}$$

Now the divergence of $\mathbf{A} \times \mathbf{B}$ is

$$\nabla \cdot (\mathbf{A} \times \mathbf{B}) = \frac{\partial}{\partial x}(A_yB_z - A_zB_y) + \frac{\partial}{\partial y}(A_zB_x - A_xB_z) + \frac{\partial}{\partial z}(A_xB_y - A_yB_x) =$$

$$\begin{aligned} &= B_z\frac{\partial A_y}{\partial x} + A_y\frac{\partial B_z}{\partial x} - B_y\frac{\partial A_z}{\partial x} - A_z\frac{\partial B_y}{\partial x} + \\ &+ B_x\frac{\partial A_z}{\partial y} + A_z\frac{\partial B_x}{\partial y} - B_z\frac{\partial A_x}{\partial y} - A_x\frac{\partial B_z}{\partial y} + \\ &+ B_y\frac{\partial A_x}{\partial y} + A_x\frac{\partial B_y}{\partial y} - B_x\frac{\partial A_y}{\partial y} - A_y\frac{\partial B_x}{\partial y} \end{aligned}$$

while

$$\mathbf{B}\cdot(\boldsymbol{\nabla}\times\mathbf{A}) = B_x\left(\frac{\partial A_z}{\partial y}-\frac{\partial A_y}{\partial z}\right)+B_y\left(\frac{\partial A_x}{\partial z}-\frac{\partial A_z}{\partial x}\right)+B_z\left(\frac{\partial A_y}{\partial x}-\frac{\partial A_x}{\partial y}\right)$$

$$\mathbf{A}\cdot(\boldsymbol{\nabla}\times\mathbf{B}) = A_x\left(\frac{\partial B_z}{\partial y}-\frac{\partial B_y}{\partial z}\right)+A_y\left(\frac{\partial B_x}{\partial z}-\frac{\partial B_z}{\partial x}\right)+A_z\left(\frac{\partial B_y}{\partial x}-\frac{\partial B_x}{\partial y}\right)$$

Taking rule (iv) and matching the terms containing $\frac{\partial}{\partial x},\frac{\partial}{\partial y},\frac{\partial}{\partial z}$:

$$\boldsymbol{\nabla}\cdot(\mathbf{A}\times\mathbf{B}) = \mathbf{B}\cdot(\boldsymbol{\nabla}\times\mathbf{A})-\mathbf{A}\cdot(\boldsymbol{\nabla}\times\mathbf{B})$$
$$\downarrow$$

$$B_z\frac{\partial A_y}{\partial x}+A_y\frac{\partial B_z}{\partial x}-B_y\frac{\partial A_z}{\partial x}-A_z\frac{\partial B_y}{\partial x} = -B_y\frac{\partial A_z}{\partial x}+B_z\frac{\partial A_y}{\partial x}-\left(-A_y\frac{\partial B_z}{\partial x}+A_z\frac{\partial B_y}{\partial x}\right)\checkmark$$

$$B_x\frac{\partial A_z}{\partial y}+A_z\frac{\partial B_x}{\partial y}-B_z\frac{\partial A_x}{\partial y}-A_x\frac{\partial B_z}{\partial y} = B_x\frac{\partial A_z}{\partial y}-B_z\frac{\partial A_x}{\partial y}-\left(A_x\frac{\partial B_z}{\partial y}-A_z\frac{\partial B_x}{\partial y}\right)\checkmark$$

$$B_y\frac{\partial A_x}{\partial y}+A_x\frac{\partial B_y}{\partial y}-B_x\frac{\partial A_y}{\partial y}-A_y\frac{\partial B_x}{\partial y} = -B_x\frac{\partial A_y}{\partial z}+B_y\frac{\partial A_x}{\partial z}-\left(-A_x\frac{\partial B_y}{\partial z}+A_y\frac{\partial B_x}{\partial z}\right)\checkmark$$

(v) Expanding the first term $\boldsymbol{\nabla}\times(f\mathbf{A})$:

$$\boldsymbol{\nabla}\times(f\mathbf{A}) = \left(\frac{\partial fA_z}{\partial y}-\frac{\partial fA_y}{\partial z}\right)\hat{\mathbf{x}}+\left(\frac{\partial fA_x}{\partial z}-\frac{\partial fA_z}{\partial x}\right)\hat{\mathbf{y}}+\left(\frac{\partial fA_y}{\partial x}-\frac{\partial fA_x}{\partial y}\right)\hat{\mathbf{z}} =$$

$$\begin{aligned} &= \left(f\frac{\partial A_z}{\partial y}+A_z\frac{\partial f}{\partial y}-f\frac{\partial A_y}{\partial z}-A_y\frac{\partial f}{\partial z}\right)\hat{\mathbf{x}}+\\ &+\left(f\frac{\partial A_x}{\partial z}+A_x\frac{\partial f}{\partial z}-f\frac{\partial A_z}{\partial x}-A_z\frac{\partial f}{\partial x}\right)\hat{\mathbf{y}}+\\ &+\left(f\frac{\partial A_y}{\partial x}+A_y\frac{\partial f}{\partial x}-f\frac{\partial A_x}{\partial y}-A_x\frac{\partial f}{\partial y}\right)\hat{\mathbf{z}}\end{aligned}$$

Expanding the second term $f(\boldsymbol{\nabla}\times\mathbf{A})$:

$$f(\boldsymbol{\nabla}\times\mathbf{A}) = f\left[\left(\frac{\partial A_z}{\partial y}-\frac{\partial A_y}{\partial z}\right)\hat{\mathbf{x}}+\left(\frac{\partial A_x}{\partial z}-\frac{\partial A_z}{\partial x}\right)\hat{\mathbf{y}}+\left(\frac{\partial A_y}{\partial x}-\frac{\partial A_x}{\partial y}\right)\hat{\mathbf{z}}\right]$$

Expanding the third term $\mathbf{A}\times(\boldsymbol{\nabla}f)$

$$\mathbf{A}\times(\boldsymbol{\nabla} f)=\mathbf{A}\times\left(\frac{\partial f}{\partial x}\hat{\mathbf{x}}+\frac{\partial f}{\partial y}\hat{\mathbf{y}}+\frac{\partial f}{\partial z}\hat{\mathbf{z}}\right)=\begin{vmatrix}\hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}}\\ A_x & A_y & A_z\\ \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} & \frac{\partial f}{\partial z}\end{vmatrix}=$$

$$=\left(A_y\frac{\partial f}{\partial z}-A_z\frac{\partial f}{\partial y}\right)\hat{\mathbf{x}}+\left(A_z\frac{\partial f}{\partial x}-A_x\frac{\partial f}{\partial z}\right)\hat{\mathbf{y}}+\left(A_x\frac{\partial f}{\partial y}-A_y\frac{\partial f}{\partial x}\right)\hat{\mathbf{z}}$$

Now match each component of both sides $\hat{\mathbf{x}},\hat{\mathbf{y}},\hat{\mathbf{z}}$:

$$\boldsymbol{\nabla}\times(f\boldsymbol{A})=f(\boldsymbol{\nabla}\times\boldsymbol{A})-\boldsymbol{A}\times(\boldsymbol{\nabla} f)$$
$$\downarrow$$

$$\left(f\frac{\partial A_z}{\partial y}+A_z\frac{\partial f}{\partial y}-f\frac{\partial A_y}{\partial z}-A_y\frac{\partial f}{\partial z}\right)\hat{\mathbf{x}}=f\left(\frac{\partial A_z}{\partial y}-\frac{\partial A_y}{\partial z}\right)\hat{\mathbf{x}}-\left(A_y\frac{\partial f}{\partial z}-A_z\frac{\partial f}{\partial y}\right)\hat{\mathbf{x}}\ \checkmark$$

$$\left(f\frac{\partial A_x}{\partial z}+A_x\frac{\partial f}{\partial z}-f\frac{\partial A_z}{\partial x}-A_y\frac{\partial f}{\partial x}\right)\hat{\mathbf{y}}=f\left(\frac{\partial A_x}{\partial z}-\frac{\partial A_z}{\partial x}\right)\hat{\mathbf{y}}-\left(A_z\frac{\partial f}{\partial x}-A_x\frac{\partial f}{\partial z}\right)\hat{\mathbf{x}}\ \checkmark$$

$$\left(f\frac{\partial A_y}{\partial x}+A_y\frac{\partial f}{\partial x}-f\frac{\partial A_x}{\partial y}-A_x\frac{\partial f}{\partial y}\right)\hat{\mathbf{z}}=f\left(\frac{\partial A_y}{\partial x}-\frac{\partial A_x}{\partial y}\right)\hat{\mathbf{z}}-\left(A_x\frac{\partial f}{\partial y}-A_y\frac{\partial f}{\partial x}\right)\hat{\mathbf{z}}\ \checkmark$$

□

Problem 1.22

$$\hat{\mathbf{r}}=\frac{\mathbf{r}}{r}=\frac{x\hat{\mathbf{x}}+y\hat{\mathbf{y}}+z\hat{\mathbf{z}}}{\sqrt{x^2+y^2+z^2}} \tag{3}$$

(a) If $\mathbf{A}$ and $\mathbf{B}$ are two vector functions, what does the expression $(\mathbf{A}\cdot\boldsymbol{\nabla})\mathbf{B}$ mean? (That is, what are its $x, y,$ and z components, in terms of the Cartesian components of $\mathbf{A}, \mathbf{B},$ and $\boldsymbol{\nabla}$?)

(b) Compute $(\hat{\mathbf{r}}\cdot\boldsymbol{\nabla})\hat{\mathbf{r}}$, where $\hat{\mathbf{r}}$ is the unit vector defined in Eq. 3

(c) For the functions in Prob. 1.15,

$$\boldsymbol{v}_a=x^2\hat{\mathbf{x}}+3xz^2\hat{\mathbf{y}}-2xz\hat{\mathbf{z}}$$
$$\boldsymbol{v}_b=xy\hat{\mathbf{x}}+2yz\hat{\mathbf{y}}+3xz\hat{\mathbf{z}}$$

evaluate $(\mathbf{v}_a\cdot\boldsymbol{\nabla})\mathbf{v}_b$

Solution:

The definition of the gradient, $\boldsymbol{\nabla}$, is

$$\boldsymbol{\nabla} = \frac{\partial}{\partial x}\hat{\mathbf{x}} + \frac{\partial}{\partial y}\hat{\mathbf{y}} + \frac{\partial}{\partial z}\hat{\mathbf{z}}$$

This gives the dot product between $\mathbf{A}$ and $\boldsymbol{\nabla}$

$$\mathbf{A} \cdot \boldsymbol{\nabla} = \left(A_x\hat{\mathbf{x}} + A_y\hat{\mathbf{y}} + A_z\hat{\mathbf{z}}\right) \cdot \left(\frac{\partial}{\partial x}\hat{\mathbf{x}} + \frac{\partial}{\partial y}\hat{\mathbf{y}} + \frac{\partial}{\partial z}\hat{\mathbf{z}}\right) =$$
$$= A_x\frac{\partial}{\partial x} + A_y\frac{\partial}{\partial y} + A_z\frac{\partial}{\partial z}$$

Now multiply by $\mathbf{B}$ on the right side

$$(\mathbf{A} \cdot \boldsymbol{\nabla})\mathbf{B} = \left(A_x\frac{\partial}{\partial x} + A_y\frac{\partial}{\partial y} + A_z\frac{\partial}{\partial z}\right)\mathbf{B} =$$
$$= A_x\frac{\partial \mathbf{B}}{\partial x} + A_y\frac{\partial \mathbf{B}}{\partial y} + A_z\frac{\partial \mathbf{B}}{\partial z}$$

What this means then, is the derivative of $\mathbf{B}$ in the direction of $\mathbf{A}$. Physically, if you think about yourself standing in a vector field $\mathbf{B}$, and taking a small step in the direction of $\mathbf{A}$, the change you experience in $\mathbf{B}$ is proportional to $(\mathbf{A} \cdot \boldsymbol{\nabla})\mathbf{B}$.

(b) Using the result from (a), we get

$$(\hat{\mathbf{r}} \cdot \boldsymbol{\nabla})\hat{\mathbf{r}} = r_x\frac{\partial \hat{\mathbf{r}}}{\partial x} + r_y\frac{\partial \hat{\mathbf{r}}}{\partial y} + r_z\frac{\partial \hat{\mathbf{r}}}{\partial z}$$

Starting with the x-component:

$$r_x\frac{\partial \hat{\mathbf{r}}}{\partial x} = \frac{x}{\sqrt{x^2+y^2+z^2}}\frac{\partial}{\partial x}\left(\frac{x\hat{\mathbf{x}} + y\hat{\mathbf{y}} + z\hat{\mathbf{z}}}{\sqrt{x^2+y^2+z^2}}\right) =$$
$$= \frac{x}{\sqrt{x^2+y^2+z^2}}\frac{\partial}{\partial x}\left(\frac{x\hat{\mathbf{x}} + y\hat{\mathbf{y}} + z\hat{\mathbf{z}}}{\sqrt{x^2+y^2+z^2}}\right)$$

Using the quotient derivative rule on the x-component and regular derivative on the y- and z-components

$$\frac{\partial}{\partial x}\left(\frac{x}{\sqrt{x^2+y^2+z^2}}\right)=\left(\frac{\sqrt{x^2+y^2+z^2}-x\frac{2x}{2\sqrt{x^2+y^2+z^2}}}{x^2+y^2+z^2}\right)=$$

$$=\frac{1}{\sqrt{x^2+y^2+z^2}}-\frac{x^2}{\sqrt{x^2+y^2+z^2}^3}=\frac{y^2+z^2}{\sqrt{x^2+y^2+z^2}^3}$$

$$\frac{\partial}{\partial x}\left(\frac{y}{\sqrt{x^2+y^2+z^2}}\right)=-\frac{xy}{\sqrt{x^2+y^2+z^2}^3}$$

$$\frac{\partial}{\partial x}\left(\frac{z}{\sqrt{x^2+y^2+z^2}}\right)=-\frac{xz}{\sqrt{x^2+y^2+z^2}^3}$$

The final form of the x-component is then

$$r_x\frac{\partial}{\partial x}\hat{\mathbf{r}}=\frac{x}{\sqrt{x^2+y^2+z^2}}\frac{y^2+z^2-xy-xz}{\sqrt{x^2+y^2+z^2}^3}$$

Doing the same calculations on the y-component gives

$$r_y\frac{\partial}{\partial y}\hat{\mathbf{r}}=\frac{y}{\sqrt{x^2+y^2+z^2}}\frac{x^2+z^2-yx-yz}{\sqrt{x^2+y^2+z^2}^3}$$

and on the z-component

$$r_z\frac{\partial}{\partial z}\hat{\mathbf{r}}=\frac{z}{\sqrt{x^2+y^2+z^2}}\frac{x^2+y^2-zx-zy}{\sqrt{x^2+y^2+z^2}^3}$$

Adding them all up gives

$$\frac{(xy^2+xz^2-x^2y-x^2z)+(yx^2+yz^2-y^2x-y^2z)+(zx^2+zy^2-z^2x-z^2y)}{(x^2+y^2+z^2)^2}=$$

$$=\frac{(xy^2-y^2x)+(xz^2-z^2x)+(yx^2-x^2y)+(yz^2-z^2y)+(zx^2-x^2z)+(zy^2-y^2z)}{(x^2+y^2+z^2)^2}=0$$

$$\downarrow$$

$$\boxed{(\hat{\mathbf{r}}\cdot\boldsymbol{\nabla})\hat{\mathbf{r}}=0}$$

(c)

$$\boldsymbol{v}_a=x^2\hat{\mathbf{x}}+3xz^2\hat{\mathbf{y}}-2xz\hat{\mathbf{z}}$$
$$\boldsymbol{v}_b=xy\hat{\mathbf{x}}+2yz\hat{\mathbf{y}}+3xz\hat{\mathbf{z}}$$

$$(\mathbf{v}_a \cdot \boldsymbol{\nabla})\mathbf{v}_b = v_{ax}\frac{\partial \mathbf{v}_b}{\partial x} + v_{ay}\frac{\partial \mathbf{v}_b}{\partial y} + v_{az}\frac{\partial \mathbf{v}_b}{\partial z} =$$
$$= x^2(y\hat{\mathbf{x}} + 3z\hat{\mathbf{z}}) + 3xz^2(x\hat{\mathbf{x}} + 2z\hat{\mathbf{y}}) - 2xz(2y\hat{\mathbf{y}} + 3x\hat{\mathbf{z}}) =$$
$$= x^2(y + 3z^2)\hat{\mathbf{x}} + 2xz(3z^2 - 2y)\hat{\mathbf{y}} - (3x^2 z)\hat{\mathbf{z}}$$

□

Problem 1.23

Problem 1.24

Problem 1.25

(ii) $\boldsymbol{\nabla}(\boldsymbol{A}\cdot\boldsymbol{B}) = \boldsymbol{A}\times(\boldsymbol{\nabla}\times\boldsymbol{B}) + \boldsymbol{B}\times(\boldsymbol{\nabla}\times\boldsymbol{A}) + (\boldsymbol{A}\cdot\boldsymbol{\nabla})\boldsymbol{B} + (\boldsymbol{B}\cdot\boldsymbol{\nabla})\boldsymbol{A}$

(iv) $\boldsymbol{\nabla}\cdot(\boldsymbol{A}\times\boldsymbol{B}) = \boldsymbol{B}\cdot(\boldsymbol{\nabla}\times\boldsymbol{A}) - \boldsymbol{A}\cdot(\boldsymbol{\nabla}\times\boldsymbol{B})$

(vi) $\boldsymbol{\nabla}\times(\boldsymbol{A}\times\boldsymbol{B}) = (\boldsymbol{B}\cdot\boldsymbol{\nabla})\boldsymbol{A} - (\boldsymbol{A}\cdot\boldsymbol{\nabla})\boldsymbol{B} + \boldsymbol{A}(\boldsymbol{\nabla}\cdot\boldsymbol{B}) - \boldsymbol{B}(\boldsymbol{\nabla}\cdot\boldsymbol{A})$

(a) Check product rule (iv) for the functions

$$\boldsymbol{A} = x\hat{\mathbf{x}} + 2y\hat{\mathbf{y}} + 3z\hat{\mathbf{z}}\ ,\quad \boldsymbol{B} = 3y\hat{\mathbf{x}} - 2x\hat{\mathbf{y}}$$

(b) Do the same for product rule (ii).

(c) Do the same for rule (vi).

Solution:

(a) Start with LHS:

$$\boldsymbol{\nabla}\cdot(\boldsymbol{A}\times\boldsymbol{B}) = \boldsymbol{\nabla}\cdot\big((A_yB_z - A_zB_y)\hat{\mathbf{x}} + (A_zB_x - A_xB_z)\hat{\mathbf{y}} + (A_xB_y - A_yB_x)\hat{\mathbf{z}}\big) =$$
$$= \frac{\partial}{\partial x}(A_yB_z - A_zB_y) + \frac{\partial}{\partial y}(A_zB_x - A_xB_z) + \frac{\partial}{\partial z}(A_xB_y - A_yB_x) =$$
$$= \frac{\partial}{\partial x}(6xz) + \frac{\partial}{\partial y}(9zy) + \frac{\partial}{\partial z}(-2x^2 - 6y^2) = 6z + 9z = \boxed{15z}$$

RHS:

$$\underbrace{\boldsymbol{B}\cdot(\boldsymbol{\nabla}\times\boldsymbol{A})}_{(1)}-\underbrace{\boldsymbol{A}\cdot(\boldsymbol{\nabla}\times\boldsymbol{B})}_{(2)}=$$

$$(1)\ \boldsymbol{B}\cdot[\boldsymbol{\nabla}\times(A)]=$$

$$=B_x\left(\frac{\partial A_z}{\partial y}-\frac{\partial A_y}{\partial z}\right)+B_y\left(\frac{\partial A_x}{\partial z}-\frac{\partial A_z}{\partial x}\right)+B_z\left(\frac{\partial A_y}{\partial x}-\frac{\partial A_x}{\partial y}\right)=0$$

$$(2)\ \boldsymbol{A}\cdot[\boldsymbol{\nabla}\times(B)]=$$

$$=A_x\left(\frac{\partial B_z}{\partial y}-\frac{\partial B_y}{\partial z}\right)+A_y\left(\frac{\partial B_x}{\partial z}-\frac{\partial B_z}{\partial x}\right)+A_z\left(\frac{\partial B_y}{\partial x}-\frac{\partial B_x}{\partial y}\right)=$$

$$=3z(-5)=-15z$$

$$(1)-(2)=0-(-15z)=\boxed{15z}$$

(b) LHS:

$$\boldsymbol{\nabla}(\boldsymbol{A}\cdot\boldsymbol{B})=\boldsymbol{\nabla}\big(A_xB_x+A_yB_y+A_zB_z\big)=\boldsymbol{\nabla}(3xy-4xy)=$$

$$=\frac{\partial(-xy)}{\partial x}\hat{\mathbf{x}}+\frac{\partial(-xy)}{\partial y}\hat{\mathbf{y}}=\boxed{-y\hat{\mathbf{x}}-x\hat{\mathbf{y}}}$$

RHS:

$$\underbrace{\boldsymbol{A}\times(\boldsymbol{\nabla}\times\boldsymbol{B})}_{(1)}+\underbrace{\boldsymbol{B}\times(\boldsymbol{\nabla}\times\boldsymbol{A})}_{(2)}+\underbrace{(\boldsymbol{A}\cdot\boldsymbol{\nabla})\boldsymbol{B}}_{(3)}+\underbrace{(\boldsymbol{B}\cdot\boldsymbol{\nabla})\boldsymbol{A}}_{(4)}$$

$$(1)\ \boldsymbol{A}\times[\boldsymbol{\nabla}\times(B)]=$$
$$=(x\hat{\mathbf{x}}+2y\hat{\mathbf{y}}+3z\hat{\mathbf{z}})\times(-5\hat{\mathbf{z}})=-10y\hat{\mathbf{x}}+5x\hat{\mathbf{y}}$$

$$(2)\ \boldsymbol{B}\times[\boldsymbol{\nabla}\times(A)]=0$$

$$(3)\ (\boldsymbol{A}\cdot\boldsymbol{\nabla})\boldsymbol{B}=$$
$$=\left(A_x\frac{\partial B_x}{\partial x}+A_y\frac{\partial B_x}{\partial y}+A_z\frac{\partial B_x}{\partial z}\right)\hat{\mathbf{x}}$$
$$+\left(A_x\frac{\partial B_y}{\partial x}+A_y\frac{\partial B_y}{\partial y}+A_z\frac{\partial B_y}{\partial z}\right)\hat{\mathbf{y}}$$
$$+\left(A_x\frac{\partial B_z}{\partial x}+A_y\frac{\partial B_z}{\partial y}+A_z\frac{\partial B_z}{\partial z}\right)\hat{\mathbf{z}}=$$
$$=6y\hat{\mathbf{x}}-2x\hat{\mathbf{y}}$$

$$(4)\ (\boldsymbol{B}\cdot\boldsymbol{\nabla})\boldsymbol{A}=$$
$$=\left(B_x\frac{\partial A_x}{\partial x}+B_y\frac{\partial A_x}{\partial y}+B_z\frac{\partial A_x}{\partial z}\right)\hat{\mathbf{x}}$$
$$+\left(B_x\frac{\partial A_y}{\partial x}+B_y\frac{\partial A_y}{\partial y}+B_z\frac{\partial A_y}{\partial z}\right)\hat{\mathbf{y}}$$
$$+\left(B_x\frac{\partial A_z}{\partial x}+B_y\frac{\partial A_z}{\partial y}+B_z\frac{\partial A_z}{\partial z}\right)\hat{\mathbf{z}}=$$
$$=3y\hat{\mathbf{x}}-4x\hat{\mathbf{y}}$$

$$(1)+(2)+(3)+(4)=(-10y+6y+3y)\hat{\mathbf{x}}+(5x-2x-4x)\hat{\mathbf{y}}=\boxed{-y\hat{\mathbf{x}}-x\hat{\mathbf{y}}}$$

(c) LHS:

$$\boldsymbol{\nabla}\times(\boldsymbol{A}\times\boldsymbol{B})=\boldsymbol{\nabla}\times\left[(A_yB_z-A_zB_y)\hat{\mathbf{x}}+(A_zB_x-A_xB_z)\hat{\mathbf{y}}+(A_xB_y-A_yB_x)\hat{\mathbf{z}}\right]=$$
$$=\boldsymbol{\nabla}\times\left[6xz\hat{\mathbf{x}}+9yz\hat{\mathbf{y}}-(2x^2+6y^2)\hat{\mathbf{z}}\right]=\boldsymbol{\nabla}\times\boldsymbol{C}=$$
$$=\boldsymbol{\nabla}\times(C)=$$
$$=(-12y-9y)\hat{\mathbf{x}}+(6x+4x)\hat{\mathbf{y}}=\boxed{-21y\hat{\mathbf{x}}+10x\hat{\mathbf{y}}}$$

RHS: From (b) it was calculated that $(\boldsymbol{A}\cdot\boldsymbol{\nabla})\boldsymbol{B}=6y\hat{\mathbf{x}}-2x\hat{\mathbf{y}}$ and $(\boldsymbol{B}\cdot\boldsymbol{\nabla})\boldsymbol{A}=3y\hat{\mathbf{x}}-4x\hat{\mathbf{y}}$. The remaining parts of (vi) RHS is

$$\boldsymbol{A}\underbrace{(\boldsymbol{\nabla}\cdot\boldsymbol{B})}_{=0}-\boldsymbol{B}\underbrace{(\boldsymbol{\nabla}\cdot\boldsymbol{A})}_{=6}=-6\boldsymbol{B}=-18y\hat{\mathbf{x}}+12x\hat{\mathbf{y}}$$

Summing it all up according to (vi), the result is

$$(3y - 6y - 18y)\hat{\mathbf{x}} + (-4x + 2x + 12x)\hat{\mathbf{y}} = \boxed{-21y\hat{\mathbf{x}} + 10x\hat{\mathbf{y}}}$$

□

Problem 1.26

Calculate the Laplacian of the following functions:

(a) $T_a = x^2 + 2xy + 3z + 4$

(b) $T_b = \sin x \sin y \sin z$

(c) $T_c = e^{-5x} \sin 4y \cos 3z$

(d) $\boldsymbol{v} = x^2\hat{\mathbf{x}} + 3xz^2\hat{\mathbf{y}} - 2xz\hat{\mathbf{z}}$

Solution:

The Laplacian of a function T is defined as

$$\boldsymbol{\nabla}^2 T = \frac{\partial T}{\partial x} + \frac{\partial T}{\partial y} + \frac{\partial T}{\partial z}$$

and the Laplacian of a vector $\boldsymbol{v}$ is defined as

$$\boldsymbol{\nabla}^2 \boldsymbol{v} = (\boldsymbol{\nabla}^2 v_x)\hat{\mathbf{x}} + (\boldsymbol{\nabla}^2 v_y)\hat{\mathbf{y}} + (\boldsymbol{\nabla}^2 v_z)\hat{\mathbf{z}}$$

Using this, the result is

(a) $\boldsymbol{\nabla}^2 T_a = 2$

(b) $\boldsymbol{\nabla}^2 T_b = -3 \sin x \sin y \sin z$

(c) $\boldsymbol{\nabla}^2 T_c = e^{-5x} \sin 4y \cos 3z(25 - 16 - 9) = 0$

(d) $\boldsymbol{\nabla}^2 \boldsymbol{v} = 2\hat{\mathbf{x}} + 6x\hat{\mathbf{y}}$

□

Problem 1.27

Prove that the divergence of a curl is always zero.

Solution:

For a vector function $\boldsymbol{v}$, show

$$\boldsymbol{\nabla} \cdot (\boldsymbol{\nabla} \times \boldsymbol{v}) = 0$$

Using the definition of the divergence and curl:

$$\boldsymbol{\nabla} \cdot \boldsymbol{v} = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} = 0$$

$$\boldsymbol{\nabla} \times \boldsymbol{v} = \left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z}\right)\hat{\mathbf{x}} + \left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x}\right)\hat{\mathbf{y}} + \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y}\right)\hat{\mathbf{z}}$$

the result is

$$\begin{aligned}\boldsymbol{\nabla} \cdot (\boldsymbol{\nabla} \times \boldsymbol{v}) &= \frac{\partial}{\partial x}\left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z}\right) + \frac{\partial}{\partial y}\left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x}\right) + \frac{\partial}{\partial z}\left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y}\right) = \\ &= \frac{\partial^2 v_z}{\partial x \partial y} - \frac{\partial^2 v_y}{\partial x \partial z} + \frac{\partial^2 v_x}{\partial y \partial z} - \frac{\partial^2 v_z}{\partial y \partial x} + \frac{\partial^2 v_y}{\partial z \partial x} - \frac{\partial^2 v_x}{\partial z \partial y} = \\ &= \underbrace{\frac{\partial^2 v_z}{\partial x \partial y} - \frac{\partial^2 v_z}{\partial y \partial x}}_{=0} + \underbrace{\frac{\partial^2 v_x}{\partial y \partial z} - \frac{\partial^2 v_x}{\partial z \partial y}}_{=0} + \underbrace{\frac{\partial^2 v_y}{\partial z \partial x} - \frac{\partial^2 v_y}{\partial x \partial z}}_{=0}\end{aligned}$$

since the symmetry of second derivatives ensures that

$$\frac{\partial^2}{\partial x \partial y} = \frac{\partial^2}{\partial y \partial x}$$

□

Problem 1.28

Prove that the curl of a gradient is always zero.

Solution:

For a function T, show

$$\boldsymbol{\nabla} \times (\boldsymbol{\nabla} T) = 0$$

Using the definitions of the gradient and curl

$$\boldsymbol{\nabla} T = \frac{\partial T}{\partial x}\hat{\mathbf{x}} + \frac{\partial T}{\partial y}\hat{\mathbf{y}} + \frac{\partial T}{\partial z}\hat{\mathbf{z}}$$

$$\boldsymbol{\nabla} \times \boldsymbol{v} = \left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z}\right)\hat{\mathbf{x}} + \left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x}\right)\hat{\mathbf{y}} + \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y}\right)\hat{\mathbf{z}}$$

results in

$$\nabla \times (\nabla T) = \underbrace{\left(\frac{\partial^2 T}{\partial y \partial z} - \frac{\partial^2 T}{\partial z \partial y}\right)}_{=0}\hat{\mathbf{x}} + \underbrace{\left(\frac{\partial^2 T}{\partial z \partial x} - \frac{\partial^2 T}{\partial x \partial z}\right)}_{=0}\hat{\mathbf{y}} + \underbrace{\left(\frac{\partial^2 T}{\partial x \partial y} - \frac{\partial^2 T}{\partial y \partial x}\right)}_{=0}\hat{\mathbf{z}}$$

since the symmetry of second derivatives ensures that

$$\frac{\partial^2}{\partial x \partial y} = \frac{\partial^2}{\partial y \partial x}$$

□

Problem 1.29

Calculate the line integral of the function $\boldsymbol{v} = x^2\hat{\mathbf{x}} + 2yz\hat{\mathbf{y}} + y^2\hat{\mathbf{z}}$ from the origin to the point $(1, 1, 1)$ by three different routes:

(a) $(0, 0, 0) \to (1, 0, 0) \to (1, 1, 0) \to (1, 1, 1)$

(b) $(0, 0, 0) \to (0, 0, 1) \to (0, 1, 1) \to (1, 1, 1)$

(c) The direct straight line

(d) What is the line integral around the closed loop that goes out along path (a) and back along path (b)?

Solution:

The line integral has the from

$$\int_a^b \boldsymbol{v} \cdot \mathrm{d}\boldsymbol{l}$$

where $\boldsymbol{dl} = dx\hat{\mathbf{x}} + dy\hat{\mathbf{y}} + dz\hat{\mathbf{z}}$

(a)

$(0, 0, 0) \to (1, 0, 0)$ has $\boldsymbol{dl} = dx\hat{\mathbf{x}}$ and x: $0 \to 1, y = 0, z = 0$.

$(1, 0, 0) \to (1, 1, 0)$ has $\boldsymbol{dl} = dy\hat{\mathbf{y}}$ and y: $0 \to 1, x = 1, z = 0$.

$(1, 1, 0) \to (1, 1, 1)$ has $\boldsymbol{dl} = dz\hat{\mathbf{z}}$ and z: $0 \to 1, x = 1, y = 1$.

$$\int_0^1 x^2 \,\mathrm{d}x + \int_0^1 2yz \,\mathrm{d}y + \int_0^1 y^2 \,\mathrm{d}z = \int_0^1 x^2 \,\mathrm{d}x + \int_0^1 1 \,\mathrm{d}z = \frac{1}{3} + 1 = \boxed{\frac{4}{3}}$$

(b)

$(0,0,0) \to (0,0,1)$ has $\mathrm{d}\boldsymbol{l} = \mathrm{d}z\hat{\mathbf{z}}$ and z: $0 \to 1, x = 0, y = 0$.

$(0,0,1) \to (0,1,1)$ has $\mathrm{d}\boldsymbol{l} = \mathrm{d}y\hat{\mathbf{y}}$ and y: $0 \to 1, x = 0, z = 1$.

$(0,1,1) \to (1,1,1)$ has $\mathrm{d}\boldsymbol{l} = \mathrm{d}x\hat{\mathbf{x}}$ and x: $0 \to 1, y = 1, z = 1$.

$$\int_0^1 y^2\,\mathrm{d}z + \int_0^1 2yz\,\mathrm{d}y + \int_0^1 x^2\,\mathrm{d}x = \int_0^1 2y\,\mathrm{d}y + \int_0^1 x^2\,\mathrm{d}x = 1 + \frac{1}{3} = \boxed{\frac{4}{3}}$$

(c) The direct straight line is obtained by setting $x = y = z = t$ and letting t: $0 \to 1$. Also, $\mathrm{d}\boldsymbol{l} = \mathrm{d}t\hat{\mathbf{t}}$, giving $\boldsymbol{v} \cdot \mathrm{d}\boldsymbol{l} = 4t^2\,\mathrm{d}t$.

$$\int_0^1 4t^2\,\mathrm{d}t = \boxed{\frac{4}{3}}$$

(d) Since the result of the line integral of $\boldsymbol{v}$ is independent of the path, going from the origin to the point $(1,1,1)$ via the path in (a), and then from the point $(1,1,1)$ to the origin via the path in (b) gives

$$\underbrace{\int_0^1 x^2\,\mathrm{d}x + \int_0^1 1\,\mathrm{d}z}_{(a)} - \left[\underbrace{\int_1^0 2y\,\mathrm{d}y + \int_1^0 x^2\,\mathrm{d}x}_{(b)\text{ in reverse}}\right] = \boxed{\frac{4}{3} - \frac{4}{3} = 0}$$

□

Problem 1.30

Calculate the surface integral of the function

$$\boldsymbol{v} = 2xz\hat{\mathbf{x}} + (x+2)\hat{\mathbf{y}} + y(z^2 - 3)\hat{\mathbf{z}}$$

over the bottom of the box. For consistency, let "upward" be the positive direction. Does the surface integral depend only on the boundary line for this function? What is the total flux over the closed surface of the box (including the bottom)?

Solution:

For the bottom of the box, x, y: $0 \to 2$, $z = 0$ and $\mathrm{d}\boldsymbol{a} = \mathrm{d}x\,\mathrm{d}y\hat{\mathbf{z}}$

$$\int_0^2\int_0^2 -3y\,\mathrm{d}x\,\mathrm{d}y = \int_0^2 \left(\int_0^2 -3y\,\mathrm{d}y\right)\mathrm{d}x = \int_0^2 -6\,\mathrm{d}x = \boxed{-12}$$

The total outward flux from all faces exluding the bottom was 20 (result from Ex. 1.7). The resulting total outward flux is then $20 - (-12) = 32$ (since the result from the bottom surface was calculated in the "wrong" direction). The surface integral does not depend only on the boundary line, since the bottom surface shares the boundary line with the combined 5 other surfaces, and the calculations gave different values (20 for the 5 surfaces, -12 for the bottom surface).

□

Problem 1.31

Calculate the volume integral of the function $T = z^2$ over the tetrahedron with corners at $(0,0,0)$, $(1,0,0)$, $(0,1,0)$, and $(0,0,1)$.

Solution:

The boundaries are

$x : 0 \to 1 - y - z$

$y : 0 \to 1 - z$

$z : 0 \to 1$

$$\begin{aligned}
&\int_0^1 z^2 \left(\int_0^{1-z} \left(\int_0^{1-y-z} \mathrm{d}x \right) \mathrm{d}y \right) \mathrm{d}z = \\
&= \int_0^1 z^2 \left(\int_0^{1-z} (1 - y - z)\, \mathrm{d}y \right) \mathrm{d}z = \\
&= \int_0^1 \frac{1}{2}(z^2 - 2z^3 + z^4)\, \mathrm{d}z \\
&= \frac{1}{2}\left(\frac{1}{3} - \frac{1}{2} + \frac{1}{5} \right) = \frac{1}{2}\left(\frac{10}{30} - \frac{15}{30} + \frac{6}{30} \right) = \frac{1}{60}
\end{aligned}$$

□

Problem 1.32

Check the fundamental theorem for gradients, using $T = x^2 + 4xy + 2yz^3$, the points $\boldsymbol{a} = (0,0,0)$, $\boldsymbol{b} = (1,1,1)$, and the three paths:

(a) $(0,0,0) \to (1,0,0) \to (1,1,0) \to (1,1,1)$

(b) $(0,0,0) \to (0,0,1) \to (0,1,1) \to (1,1,1)$

(c) the parabolic path $z = x^2$, $y = x$

Solution:

The fundamental theorem for gradients says

$$\int_a^b (\boldsymbol{\nabla} T) \cdot \mathrm{d}\boldsymbol{l} = T(\boldsymbol{b}) - T(\boldsymbol{a})$$

$$\boldsymbol{\nabla} T = (2x + 4y)\hat{\mathbf{x}} + (4x + 2z^3)\hat{\mathbf{y}} + (6yz^2)\hat{\mathbf{z}}$$

RHS for all paths is

$$T(\boldsymbol{b}) - T(\boldsymbol{a}) = 7 - 0 = \boxed{7}$$

which is what we want to check for the LHS of the different paths

(a)

$(0,0,0) \to (1,0,0)$ has $x : 0 \to 1$, $y = z = 0$ and $\mathrm{d}\boldsymbol{l} = \mathrm{d}x\hat{\mathbf{x}}$

$$\text{(i)} \int_0^1 2x \, \mathrm{d}x = 1$$

$(1,0,0) \to (1,1,0)$ has $y : 0 \to 1$, $x = 1$, $z = 0$ and $\mathrm{d}\boldsymbol{l} = \mathrm{d}y\hat{\mathbf{y}}$

$$\text{(ii)} \int_0^1 4 \, \mathrm{d}y = 4$$

$(1,1,0) \to (1,1,1)$ has $z : 0 \to 1$, $x = 1$, $y = 1$ and $\mathrm{d}\boldsymbol{l} = \mathrm{d}z\hat{\mathbf{z}}$

$$\text{(iii)} \int_0^1 6z^2 \, \mathrm{d}z = 2$$

$$\text{(i)} + \text{(ii)} + \text{(iii)} = 1 + 4 + 2 = \boxed{7}$$

(b)

$(0,0,0) \to (0,0,1)$ has $z : 0 \to 1$, $x = y = 0$ and $\mathrm{d}\boldsymbol{l} = \mathrm{d}z\hat{\mathbf{z}}$

$$\text{(i)} \int_0^1 0 \, \mathrm{d}y = 0$$

$(0,0,1) \to (0,1,1)$ has $y : 0 \to 1$, $x = 0$, $z = 1$ and $\mathrm{d}\boldsymbol{l} = \mathrm{d}y\hat{\mathbf{y}}$

$$\text{(ii)} \int_0^1 2 \, \mathrm{d}y = 2$$

$(0,1,1) \to (1,1,1)$ has $x : 0 \to 1$, $y = 1$, $z = 1$ and $\mathrm{d}\boldsymbol{l} = \mathrm{d}x\hat{\mathbf{x}}$

$$\text{(iii)} \int_0^1 2x + 4\,\mathrm{d}x = 5$$

$$\text{(i)} + \text{(ii)} + \text{(iii)} = 0 + 2 + 5 = \boxed{7}$$

(c) Parametrization $x = y = t,\ z = t^2$ gives $\boldsymbol{r}(t) = (t, t, t^2)$, $t : 0 \to 1$. This means $\mathrm{d}\boldsymbol{l} = \boldsymbol{r}'(t)\,\mathrm{d}t = (1, 1, 2t)\,\mathrm{d}t$

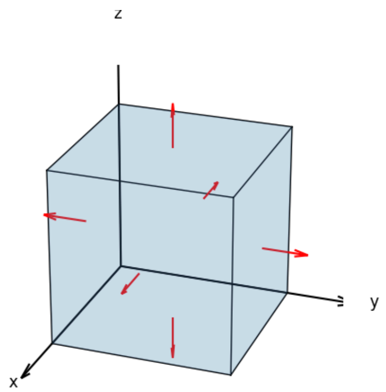
$$\int_0^1 (6t, 4t + 2t^6, 6t^5) \cdot (1, 1, 2t)\,\mathrm{d}t = \int_0^1 (6t + 4t + 2t^6 + 12t^6)\,\mathrm{d}t =$$

$$= \int_0^1 10t + 2t^6 + 12t^6\,\mathrm{d}t = 5 + \frac{2}{7} + \frac{12}{7} = \boxed{7}$$

□

Problem 1.33

Test the divergence theorem for the function $\boldsymbol{v} = (xy)\hat{\mathbf{x}} + (2yz)\hat{\mathbf{y}} + (3xz)\hat{\mathbf{z}}$. Take as your volume the cube with corners $(0,0,0)$,$(0,0,2)$,$(0,2,0)$,$(2,0,0)$



Solution:

The divergence theorem says

$$\int_V (\boldsymbol{\nabla} \cdot \boldsymbol{v})\,\mathrm{d}\tau = \oint_S \boldsymbol{v} \cdot \mathrm{d}\boldsymbol{a}$$

The divergence of $\boldsymbol{v}$ is

$$\boldsymbol{\nabla} \cdot \boldsymbol{v} = y + 2z + 3x$$

and the resulting LHS is

$$\iiint_0^2 (y+2z+3x)\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z = \iint_0^2 \left[yx + 2zx + \frac{3x^2}{2}\right]_0^2 \mathrm{d}y\,\mathrm{d}z =$$

$$= \iint_0^2 2y + 4z + 6\,\mathrm{d}y\,\mathrm{d}z = \int_0^2 \left[y^2 + 4yz + 6y\right]_0^2 \mathrm{d}z =$$

$$= \int_0^2 16 + 8z\,\mathrm{d}z = \left[16z + 4z^2\right]_0^2 = \boxed{48}$$

Top face:

$$x,y : 0 \to 2\ ,\ z = 2\ , \mathrm{d}\boldsymbol{a} = \mathrm{d}x\,\mathrm{d}y\hat{\mathbf{z}}$$

$$\int_S 6x\,\mathrm{d}x\,\mathrm{d}y = 24$$

Bottom face:

$$x,y : 0 \to 2\ ,\ z = 0\ , \mathrm{d}\boldsymbol{a} = \mathrm{d}x\,\mathrm{d}y(-\hat{\mathbf{z}})$$

$$\int_S 0\,\mathrm{d}x\,\mathrm{d}y = 0$$

Left face:

$$x,z : 0 \to 2\ ,\ y = 0\ , \mathrm{d}\boldsymbol{a} = \mathrm{d}x\,\mathrm{d}z(-\hat{\mathbf{y}})$$

$$\int_S 0\,\mathrm{d}x\,\mathrm{d}z = 0$$

Right face:

$$x,z : 0 \to 2\ ,\ y = 2\ , \mathrm{d}\boldsymbol{a} = \mathrm{d}x\,\mathrm{d}z\hat{\mathbf{y}}$$

$$\int_S 4z\,\mathrm{d}x\,\mathrm{d}z = 16$$

Front face:

$$y,z : 0 \to 2\ ,\ x = 2\ , \mathrm{d}\boldsymbol{a} = \mathrm{d}y\,\mathrm{d}z\hat{\mathbf{x}}$$

$$\int_S 2y\,\mathrm{d}y\,\mathrm{d}z = 8$$

Back face:

$$y,z : 0 \to 2\ ,\ x = 0\ , \mathrm{d}\boldsymbol{a} = \mathrm{d}y\,\mathrm{d}z(-\hat{\mathbf{x}})$$

$$\int_S 0\,\mathrm{d}y\,\mathrm{d}z = 0$$

Total is $\boxed{48}$

$\square$

Problem 1.34

Test Stokes' theorem for the function $\boldsymbol{v} = (xy)\hat{\mathbf{x}} + (2yz)\hat{\mathbf{y}} + (3xz)\hat{\mathbf{z}}$, using the triangular area with points $(0,0,0)$, $(0,1,0)$,$(0,0,1)$.

Solution:

Stokes's theorem

$$\int_S (\boldsymbol{\nabla} \times \boldsymbol{v}) \cdot \mathrm{d}\boldsymbol{a} = \oint_P \boldsymbol{v} \cdot \mathrm{d}\boldsymbol{l}$$

$$\boldsymbol{\nabla} \times \boldsymbol{v} = (-2y)\hat{\mathbf{x}} + (-3z)\hat{\mathbf{y}} + (-x)\hat{\mathbf{z}}$$

$$\mathrm{d}\boldsymbol{a} = \mathrm{d}y\,\mathrm{d}z\hat{\mathbf{x}}$$

$$y : 0 \to 2 - z$$

$$z : 0 \to 2$$

$$\int_0^2 \left(\int_0^{2-z} -2y\,\mathrm{d}y \right) \mathrm{d}z =$$

$$= \int_0^2 \left[-y^2\right]_0^{2-z} \mathrm{d}z = \int_0^2 (-4 + 4z - z^2) = -8 + 8 - \frac{8}{3} = \boxed{-\frac{8}{3}}$$

Line 1:

$$(0,0,0) \to (0,2,0)$$

$$\mathrm{d}\boldsymbol{l} = \mathrm{d}x\hat{\mathbf{x}}$$

$$x = z = 0$$

$$\int_0^2 0\,\mathrm{d}x = 0$$

Line 2:

$$(0,2,0) \to (0,0,2)$$

$$\boldsymbol{r}(t) = (0, 2-t, t),\ t : 0 \to 2$$

$$\mathrm{d}\boldsymbol{l} = \boldsymbol{r}'(t)\,\mathrm{d}t$$

$$\int_0^2 (0, 2(2-t)(t), 0) \cdot (0, -1, 1)\,\mathrm{d}t = \int_0^2 -4t + 2t^2\,\mathrm{d}t = -2t^2 + \frac{2}{3}t^3 = -8 + \frac{16}{3} = -\frac{8}{3}$$

Line 3:

$$(0,0,2) \to (0,0,0)$$
$$\mathrm{d}\boldsymbol{l} = \mathrm{d}z\hat{\mathbf{z}}$$
$$y = x = 0$$
$$\int_2^0 0\,\mathrm{d}z = 0$$

All line integrals combined gives $\boxed{-\frac{8}{3}}$

□

Problem 1.35

(a) Show that

$$\int_S f(\boldsymbol{\nabla} \times \boldsymbol{A}) \cdot \mathrm{d}\boldsymbol{a} = \int_S [\boldsymbol{A} \times (\boldsymbol{\nabla} f)] \cdot \mathrm{d}\boldsymbol{a} + \oint_P f\boldsymbol{A} \cdot \mathrm{d}\boldsymbol{l}$$

(b) Show that

$$\int_V \boldsymbol{B} \cdot (\boldsymbol{\nabla} \times \boldsymbol{A})\,\mathrm{d}\tau = \int_V \boldsymbol{A} \cdot (\boldsymbol{\nabla} \times \boldsymbol{B})\,\mathrm{d}\tau + \oint_S (\boldsymbol{A} \times \boldsymbol{B}) \cdot \mathrm{d}\boldsymbol{a}$$

Solution:

(a) Using product rule (7) (from the cover of the book)

$$(7)\ \boldsymbol{\nabla} \times (f\boldsymbol{A}) = f(\boldsymbol{\nabla} \times \boldsymbol{A}) - \boldsymbol{A} \times (\boldsymbol{\nabla} f)$$
$$\downarrow$$
$$f(\boldsymbol{\nabla} \times \boldsymbol{A}) = \boldsymbol{A} \times (\boldsymbol{\nabla} f) + \boldsymbol{\nabla} \times (f\boldsymbol{A})$$

Integrating this expression over a surface gives

$$\int_S f(\boldsymbol{\nabla} \times \boldsymbol{A}) \cdot \mathrm{d}\boldsymbol{a} = \int_S [\boldsymbol{A} \times (\boldsymbol{\nabla} f)] \cdot \mathrm{d}\boldsymbol{a} + \underbrace{\int_S \boldsymbol{\nabla} \times (f\boldsymbol{A}) \cdot \mathrm{d}\boldsymbol{a}}_{\text{use Stokes' Theorem}}$$
$$\downarrow$$
$$\int_S f(\boldsymbol{\nabla} \times \boldsymbol{A}) \cdot \mathrm{d}\boldsymbol{a} = \int_S [\boldsymbol{A} \times (\boldsymbol{\nabla} f)] \cdot \mathrm{d}\boldsymbol{a} + \oint_P f\boldsymbol{A} \cdot \mathrm{d}\boldsymbol{l}$$

(b) Using product rule (6)

$$\boldsymbol{\nabla}\cdot(\boldsymbol{A}\times\boldsymbol{B}) = \boldsymbol{B}\cdot(\boldsymbol{\nabla}\times\boldsymbol{A}) - \boldsymbol{A}\cdot(\boldsymbol{\nabla}\times\boldsymbol{B})$$
$$\downarrow$$
$$\boldsymbol{B}\cdot(\boldsymbol{\nabla}\times\boldsymbol{A}) = \boldsymbol{A}\cdot(\boldsymbol{\nabla}\times\boldsymbol{B}) + \boldsymbol{\nabla}\cdot(\boldsymbol{A}\times\boldsymbol{B})$$

Integrating this expression over a volume gives

$$\int_V \boldsymbol{B}\cdot(\boldsymbol{\nabla}\times\boldsymbol{A})\,\mathrm{d}\tau = \int_V \boldsymbol{A}\cdot(\boldsymbol{\nabla}\times\boldsymbol{B})\,\mathrm{d}\tau + \underbrace{\int_V \boldsymbol{\nabla}\cdot(\boldsymbol{A}\times\boldsymbol{B})\,\mathrm{d}\tau}_{\text{use Gauss' Theorem}}$$
$$\downarrow$$
$$\int_V \boldsymbol{B}\cdot(\boldsymbol{\nabla}\times\boldsymbol{A})\,\mathrm{d}\tau = \int_V \boldsymbol{A}\cdot(\boldsymbol{\nabla}\times\boldsymbol{B})\,\mathrm{d}\tau + \oint_S (\boldsymbol{A}\times\boldsymbol{B})\cdot\mathrm{d}\boldsymbol{a}$$

□

Problem 1.36

(a) Show that

$$\int_{\mathcal{S}} f(\boldsymbol{\nabla}\times\mathbf{A})\cdot\mathrm{d}\mathbf{a} = \int_{\mathcal{S}}[\mathbf{A}\times(\boldsymbol{\nabla}f)]\cdot\mathrm{d}\mathbf{a} + \oint_{\mathcal{P}} f\mathbf{A}\cdot\mathrm{d}\mathbf{l}$$

(b) Show that

$$\int_{\mathcal{V}} \mathbf{B}\cdot(\boldsymbol{\nabla}\times\mathbf{A})\,\mathrm{d}\tau = \int_{\mathcal{V}} \mathbf{A}\cdot(\boldsymbol{\nabla}\times\mathbf{B})\,\mathrm{d}\tau + \oint_{\mathcal{S}}(\mathbf{A}\times\mathbf{B})\cdot\mathrm{d}\mathbf{a}$$

Solution:

(a) Using product rule (v):

$$\text{(v)}\ \boldsymbol{\nabla}\times(f\mathbf{A}) = f(\boldsymbol{\nabla}\times\mathbf{A}) - \mathbf{A}\times(\boldsymbol{\nabla}f)$$
$$\downarrow$$
$$f(\boldsymbol{\nabla}\times\mathbf{A}) = \mathbf{A}\times(\boldsymbol{\nabla}f) + \boldsymbol{\nabla}\times(f\mathbf{A})$$

Integrating this expression over a surface $\mathcal{S}$ yields

$$\int_{\mathcal{S}} f(\boldsymbol{\nabla}\times\mathbf{A})\cdot\mathrm{d}\mathbf{a} = \int_{\mathcal{S}} \mathbf{A}\times(\boldsymbol{\nabla}f)\cdot\mathrm{d}\mathbf{a} + \int_{\mathcal{S}} \boldsymbol{\nabla}\times(f\mathbf{A})\cdot\mathrm{d}\mathbf{a}\ .$$

Using Stokes' Theoreom on the last part gives the answer

$$\boxed{\int_{\mathcal{S}} f(\boldsymbol{\nabla}\times\mathbf{A})\cdot\mathrm{d}\mathbf{a} = \int_{\mathcal{S}}[\mathbf{A}\times(\boldsymbol{\nabla}f)]\cdot\mathrm{d}\mathbf{a} + \oint_{\mathcal{P}} f\mathbf{A}\cdot\mathrm{d}\mathbf{l}}$$

(b) Using product rule (iv)

$$\text{(iv)}\ \boldsymbol{\nabla}\cdot(\mathbf{A}\times\mathbf{B}) = \mathbf{B}\cdot(\boldsymbol{\nabla}\times\mathbf{A}) - \mathbf{A}\cdot(\boldsymbol{\nabla}\times\mathbf{B})$$
$$\downarrow$$
$$\mathbf{B}\cdot(\boldsymbol{\nabla}\times\mathbf{A}) = \mathbf{A}\cdot(\boldsymbol{\nabla}\times\mathbf{B}) + \boldsymbol{\nabla}\cdot(\mathbf{A}\times\mathbf{B})$$

Integrating this expression over a volume $\mathcal{V}$ yields

$$\int_{\mathcal{V}} \mathbf{B}\cdot(\boldsymbol{\nabla}\times\mathbf{A})\,\mathrm{d}\tau = \int_{\mathcal{V}} \mathbf{A}\cdot(\boldsymbol{\nabla}\times\mathbf{B})\,\mathrm{d}\tau + \int_{\mathcal{V}} \boldsymbol{\nabla}\cdot(\mathbf{A}\times\mathbf{B})\,\mathrm{d}\tau\ .$$

Using Gauss' Theorem / Divergence Theorem on the last part gives the answer

$$\boxed{\int_{\mathcal{V}} \mathbf{B}\cdot(\boldsymbol{\nabla}\times\mathbf{A})\,\mathrm{d}\tau = \int_{\mathcal{V}} \mathbf{A}\cdot(\boldsymbol{\nabla}\times\mathbf{B})\,\mathrm{d}\tau + \oint_{\mathcal{S}} (\mathbf{A}\times\mathbf{B})\cdot\mathrm{d}\mathbf{a}}$$

□

Problem 1.37

Find formulas for r, θ, ϕ in terms of x, y, z.

Solution:

In spherical coordinates we have

$$\begin{aligned} x &= r\sin\theta\cos\phi \\ y &= r\sin\theta\sin\phi \\ z &= r\cos\theta \end{aligned}$$

***r*-component:**

From Pythagoras we know that $r = \sqrt{x^2+y^2+z^2} \longleftrightarrow r^2 = x^2+y^2+z^2$. This leads us to start with the r-component, by squaring each relation and adding them up:

$$\begin{aligned} x^2 &= r^2\sin^2\theta\cos^2\phi \\ y^2 &= r^2\sin^2\theta\sin^2\phi \\ z^2 &= r^2\cos^2\theta \end{aligned}$$
$$\downarrow$$
$$x^2+y^2+z^2 = r^2\big(\sin^2\theta\cos^2\phi + \sin^2\theta\sin^2\phi + \cos^2\theta\big)$$

All the trigonometric functions above should then be equal to one, which is confirmed by rearranging and simplifying

$$= r^2 \left(\sin^2\theta \underbrace{(\cos^2\phi + \sin^2\phi)}_{=1} + \cos^2\theta \right) = r^2 \underbrace{(\sin^2\theta + \cos^2\theta)}_{=1} = r^2$$

$$\downarrow$$

$$\boxed{r = \sqrt{x^2 + y^2 + z^2}}$$

***θ*-component**:

Since θ is alone in the relation for z, we can use z to solve for θ. Rewriting the term for z leads to

$$z = r\cos\theta \to \cos\theta = \frac{z}{r} = \frac{z}{\sqrt{x^2 + y^2 + z^2}}$$

$$\downarrow$$

$$\boxed{\theta = \arccos\left(\frac{z}{\sqrt{x^2 + y^2 + z^2}}\right)}$$

***ϕ*-component:** The x and y components both share a factor of $r\sin\theta$, which makes the ϕ component appear by itself if we divide y by x (or x by y)

$$\frac{y}{x} = \frac{r\sin\theta\sin\phi}{r\sin\theta\cos\phi} = \frac{\sin\phi}{\cos\phi} = \tan\phi$$

$$\downarrow$$

$$\boxed{\phi = \arctan\left(\frac{y}{x}\right)}$$

The relations for r, θ, ϕ in terms of x, y, z are then

$$\boxed{\begin{aligned} & r = \sqrt{x^2 + y^2 + z^2} \\ & \theta = \arccos\left(\frac{z}{\sqrt{x^2 + y^2 + z^2}}\right) \\ & \phi = \arctan\left(\frac{y}{x}\right) \end{aligned}}$$

□

Problem 1.38

Express the unit vectors $\hat{\mathbf{r}}, \hat{\boldsymbol{\theta}}, \hat{\boldsymbol{\phi}}$ in terms of $\hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}}$. Check your answers several ways ($\hat{\mathbf{r}} \cdot \hat{\mathbf{r}} \stackrel{?}{=} 1, \hat{\boldsymbol{\theta}} \cdot \hat{\boldsymbol{\phi}} \stackrel{?}{=} 0, \hat{\mathbf{r}} \times \hat{\boldsymbol{\theta}} \stackrel{?}{=} \hat{\boldsymbol{\phi}}, \ldots$). Also work out the inverse formulas, giving $\hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}}$ in terms of $\hat{\mathbf{r}}, \hat{\boldsymbol{\theta}}, \hat{\boldsymbol{\phi}}$ (and ϕ, θ).

Problem 1.39

(a) Check the divergence theorem for the function $\mathbf{v}_1 = r^2\hat{\mathbf{r}}$, using as your volume the sphere of radius R, centered at the origin.

(b) Do the same for $\mathbf{v}_2 = (1/r^2)\hat{\mathbf{r}}$.

Solution:

The divergence theorem is

$$\int_{\mathcal{V}} \boldsymbol{\nabla} \cdot r^2\hat{\mathbf{r}}\, \mathrm{d}\tau = \oint_{\mathcal{S}} r^2\hat{\mathbf{r}} \cdot \mathrm{d}\mathbf{a}$$

Taking the divergence of $\mathbf{v}_1$ in spherical coordinates gives

$$\boldsymbol{\nabla} \cdot r^2\hat{\mathbf{r}} = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 v_{1r}) = \frac{1}{r^2}\frac{\partial r^4}{\partial r} = \frac{4r^3}{r^2} = 4r$$

In spherical coordinates,

$$\mathrm{d}\tau = r^2 \sin\theta\, \mathrm{d}r\, \mathrm{d}\theta\, \mathrm{d}\phi$$
$$\mathrm{d}\mathbf{a} = r^2 \sin\theta\, \mathrm{d}\theta\, \mathrm{d}\phi\hat{\mathbf{r}}$$

LHS of the divergence theorem becomes

$$\int_{\mathcal{V}} 4rr^2 \sin\theta\, \mathrm{d}r\, \mathrm{d}\theta\, \mathrm{d}\phi = 4\int_0^R r^3 \int_0^\pi \sin\theta \int_0^{2\pi} \mathrm{d}r\, \mathrm{d}\theta\, \mathrm{d}\phi =$$
$$= 8\pi \int_0^R r^3 \int_0^\pi \sin\theta\, \mathrm{d}r\, \mathrm{d}\theta = 16\pi \int_0^R r^3\, \mathrm{d}r = \boxed{4\pi R^4}$$

and RHS

$$\oint_{\mathcal{S}} r^2 r^2 \sin\theta\, \mathrm{d}\theta\, \mathrm{d}\phi = r^4 \int_0^\pi \sin\theta \int_0^{2\pi} \mathrm{d}\theta\, \mathrm{d}\phi = 2\pi r^4 \int_0^\pi \sin\theta\, \mathrm{d}\theta = 4\pi r^4\big|_{r=R} = \boxed{4\pi R^4}$$

which confirms the divergence theorem for $\mathbf{v}_1$.

(b)

The divergence of $\mathbf{v}_2$ is

$$\nabla \cdot \mathbf{v}_2 = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 v_{2r}) = \frac{1}{r^2}\frac{\partial 1}{\partial r} = \boxed{0}$$

so LHS of divergence theorem is zero (which is not completely true, it's zero everywhere except at $r = 0$, so the calculation here is not fully correct). The RHS is

$$\oint_{\mathcal{S}} = \frac{1}{r^2} r^2 \sin\theta \, \mathrm{d}\theta \mathrm{d}\phi = \int_0^{\pi} \sin\theta \int_0^{2\pi} \mathrm{d}\theta \mathrm{d}\phi = 2\pi \int_0^{\pi} \sin\theta \, \mathrm{d}\theta = \boxed{4\pi}$$

□

Problem 1.40

Compute the divergence of

$$\mathbf{v} = (r\cos\theta)\hat{\mathbf{r}} + (r\sin\theta)\hat{\boldsymbol{\theta}} + (r\sin\theta\cos\phi)\hat{\boldsymbol{\phi}}$$

Check the divergence theorem for this function, using as your volume the inverted hemispherical bowl of radius R, resting on the xy plane and centered at the origin.

Solution:

The divergence of $\mathbf{v}$ is

$$\begin{aligned}\nabla \cdot \mathbf{v} &= \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 v_r) + \frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}(\sin\theta v_\theta) + \frac{1}{r\sin\theta}\frac{\partial v_\phi}{\partial\phi} = \\ &= 3\cos\theta + 2\cos\theta - \sin\phi = 5\cos\theta - \sin\phi\end{aligned}$$

The divergence theorem is

$$\int_{\mathcal{V}} \nabla \cdot \mathbf{v} \, \mathrm{d}\tau = \oint_{\mathcal{S}} \mathbf{v} \cdot \mathrm{d}\mathbf{a}$$

The volume integral is

$$\int_0^R \int_0^{\frac{\pi}{2}} \int_0^{2\pi} (5\cos\theta - \sin\phi) r^2 \sin\theta \, \mathrm{d}r \, \mathrm{d}\theta \, \mathrm{d}\phi =$$

$$= \int_0^R r^2 \left(\int_0^{\frac{\pi}{2}} \sin\theta \left(\int_0^{2\pi} (5\cos\theta - \sin\phi) \, \mathrm{d}\phi \right) \mathrm{d}\theta \right) \mathrm{d}r =$$

$$= \int_0^R r^2 \left(\int_0^{\frac{\pi}{2}} \sin\theta ([5\cos\theta\phi + \cos\phi]_0^{2\pi}) \, \mathrm{d}\theta \right) \mathrm{d}r =$$

$$= \int_0^R r^2 \left(\int_0^{\frac{\pi}{2}} \sin\theta (10\pi\cos\theta + 1 - 1) \, \mathrm{d}\theta \right) \mathrm{d}r =$$

$$= \int_0^R r^2 \left(\int_0^{\frac{\pi}{2}} 10\pi \sin\theta \cos\theta \, \mathrm{d}\theta \right) \mathrm{d}r =$$

$$= \int_0^R r^2 \left(\int_0^{\frac{\pi}{2}} 5\pi \sin 2\theta \, \mathrm{d}\theta \right) \mathrm{d}r =$$

$$= \int_0^R r^2 \left[\frac{-5\pi\cos 2\theta}{2} \right]_0^{\frac{\pi}{2}} \mathrm{d}r = \int_0^R r^2 \left[\frac{5\pi}{2} - \frac{-5\pi}{2} \right] \mathrm{d}r = \int_0^R 5\pi r^2 \, \mathrm{d}r =$$

$$= \boxed{\frac{5\pi R^3}{3}}$$

The closed surface integral is split into the circular disc ($\mathcal{S}_1$) and the half surface of the sphere ($\mathcal{S}_2$)

$$\oint_{\mathcal{S}} = \int_{\mathcal{S}_1} + \int_{\mathcal{S}_2}$$

For $\mathcal{S}_1$:

$$\mathbf{da} = \hat{\boldsymbol{\theta}} r \, \mathrm{d}r \, \mathrm{d}\phi, \quad \theta = \frac{\pi}{2}, \quad r : 0 \to R, \quad \phi : 0 \to 2\pi$$

so the integral becomes

$$\int_0^R \int_0^{2\pi} r^2 \sin\theta \, \mathrm{d}r \, \mathrm{d}\phi = \int_0^R r^2 \left(\int_0^{2\pi} \mathrm{d}\phi \right) \mathrm{d}r = 2\pi \int_0^R r^2 \, \mathrm{d}r = \frac{2\pi R^3}{3}$$

For $\mathcal{S}_2$:

$$\mathbf{da} = \hat{\mathbf{r}} r^2 \sin\theta \, \mathrm{d}\theta \, \mathrm{d}\phi, \quad r = R, \quad \theta : 0 \to \frac{\pi}{2}, \quad \phi : 0 \to 2\pi$$

so the integral becomes

$$\int_0^{\frac{\pi}{2}} \int_0^{2\pi} r\cos\theta r^2 \sin\theta \, \mathrm{d}\theta \mathrm{d}\phi = R^3 \int_0^{\frac{\pi}{2}} \sin\theta\cos\theta \left(\int_0^{2\pi} \mathrm{d}\phi \right) \mathrm{d}\theta =$$

$$= 2\pi R^3 \int_0^{\frac{\pi}{2}} \sin\theta\cos\theta \, \mathrm{d}\theta = 2\pi R^3 \int_0^{\frac{\pi}{2}} \frac{1}{2} \sin 2\theta \, \mathrm{d}\theta =$$

$$= 2\pi R^3 \left[-\frac{\cos 2\theta}{4} \right]_0^{\frac{\pi}{2}} = 2\pi R^3 \left(\frac{1}{4} - \frac{-1}{4} \right) = \pi R^3 = \frac{3\pi R^3}{3}$$

Adding these results we get

$$\frac{2\pi R^3}{3} + \frac{3\pi R^3}{3} = \boxed{\frac{5\pi R^3}{3}}$$

which is the same result we got from the volume integral.

□

Problem 1.41

Problem 1.42

Problem 1.43

Problem 1.44

Evaluate the following integrals:

(a) $\int_2^6 (3x^2 - 2x - 1)\delta(x-3) \, \mathrm{d}x$

(b) $\int_0^5 \cos(x)\delta(x-\pi) \, \mathrm{d}x$

(c) $\int_0^3 x^3 \delta(x+1) \, \mathrm{d}x$

(d) $\int_{-\infty}^{\infty} \ln(x+3)\delta(x+2) \, \mathrm{d}x$

Solution:

Integral of delta-dirac

$$\int_a^b \delta(x-p)\,\mathrm{d}x = \begin{cases} 1, \text{ if } p \in (a,b) \\ 0, \text{ otherwise} \end{cases}$$

(a) Zero everywhere except at $x = 3$ which gives

$$20 \underbrace{\int_2^6 \delta(x-3)\,\mathrm{d}x}_{=1 \text{ at } x=3} = 20$$

(b) Zero everywhere except at $x = \pi$ which gives

$$-1 \int_0^5 \delta(x-\pi)\,\mathrm{d}x = -1$$

(c) Zero everywhere except at $x = -1$, which is not included in the integral bounds, therefore it is zero.

(d) Zero everywhere except at $x = -2$ which gives

$$\ln(1) \int_{-\infty}^{\infty} \delta(x+2)\,\mathrm{d}x = 0$$

□

Problem 1.45

Evaluate the following integrals:

(a) $\int_{-2}^{2}(2x+3)\delta(3x)\,\mathrm{d}x$

(b) $\int_0^2 (x^3+3x+2)\delta(1-x)\,\mathrm{d}x$

(c) $\int_{-1}^{1} 9x^2\delta(3x+1)\,\mathrm{d}x$

(d) $\int_{-\infty}^{a} \delta(x-b)\,\mathrm{d}x$

Solution:

For the following problems we use

$$\delta(kx) = \frac{1}{|k|}\delta(x)$$

(a)

$$\int_{-2}^{2}(2x+3)\delta(3x)\,\mathrm{d}x = 3\int_{-2}^{2}\frac{1}{3}\delta(x)\,\mathrm{d}x = 1$$

(b)

$$\int_0^2 (x^3+3x+2)\delta(1-x)\,\mathrm{d}x = \int_0^2 (x^3+3x+2)\delta(x-1)\,\mathrm{d}x = 6\int_0^2 \delta(x-1)\,\mathrm{d}x = 6$$

(c)

$$\delta(3x+1) = \frac{1}{3}\delta\left(x+\frac{1}{3}\right)$$
$$\downarrow$$
$$\int_{-1}^{1} 9x^2\delta(3x+1)\,\mathrm{d}x = \int_{-1}^{1} \frac{1}{3}\delta\left(x+\frac{1}{3}\right)\mathrm{d}x = \frac{1}{3}$$

(d)

$$\int_{-\infty}^{a} \delta(x-b)\,\mathrm{d}x = \begin{cases} 1, \text{ if } a>b \\ 0, \text{ if } a<b \end{cases}$$

□

Problem 1.46

(a) Show that

$$x\frac{\mathrm{d}}{\mathrm{d}x}(\delta(x)) = -\delta(x)$$

(b) Let $\theta(x)$ be the **step function**:

$$\theta(x) = \begin{cases} 1, & \text{if} > 0 \\ 0, & \text{if } x \leq 0 \end{cases}$$

Show that $\mathrm{d}\theta/\,\mathrm{d}x = \delta(x)$

Solution:

(a)

$$\int_{-\infty}^{\infty} x\frac{\mathrm{d}}{\mathrm{d}x}(\delta(x))\,\mathrm{d}x = \underbrace{[x\delta(x)]_{-\infty}^{\infty}}_{=0} - \int_{-\infty}^{\infty} \delta(x)\,\mathrm{d}x$$
$$\downarrow$$
$$\int_{-\infty}^{\infty} x\frac{\mathrm{d}}{\mathrm{d}x}(\delta(x))\,\mathrm{d}x = -\int_{-\infty}^{\infty} \delta(x)\,\mathrm{d}x \rightarrow \boxed{x\frac{\mathrm{d}}{\mathrm{d}x}(\delta(x)) = -\delta(x)}$$

(b) Using test function $f(x)$ and integrating:

$$f(x)\frac{\mathrm{d}\theta(x)}{\mathrm{d}x} \to \int_{-\infty}^{\infty} f(x)\frac{\mathrm{d}\theta(x)}{\mathrm{d}x}\,\mathrm{d}x = \underbrace{[f(x)\theta(x)]_{-\infty}^{\infty}}_{=f(\infty)} - \int_{-\infty}^{\infty} f'(x)\theta(x)\,\mathrm{d}x =$$

$$= f(\infty) - \int_{0}^{\infty} f'(x)\,\mathrm{d}x = f(\infty) - f(\infty) + f(0) = f(0) = f(x)\delta(x)$$

$$\downarrow$$

$$f(x)\frac{\mathrm{d}\theta(x)}{\mathrm{d}x} = f(x)\delta(x) \to \boxed{\frac{\mathrm{d}\theta(x)}{\mathrm{d}x} = \delta(x)}$$

□

Problem 1.47

Problem 1.48

Problem 1.49

Problem 1.50

(a) Let $\boldsymbol{F}_1 = x^2\hat{\mathbf{z}}$ and $\boldsymbol{F}_2 = x\hat{\mathbf{x}} + y\hat{\mathbf{y}} + z\hat{\mathbf{z}}$. Calculate the divergence and curl of $\boldsymbol{F}_1$ and $\boldsymbol{F}_2$. Which one can be written as the gradient of a scalar? Find a scalar potential that does the job. Which one can be written as the curl of a vector? Find a suitable vector potential.

(b) Show that $\boldsymbol{F}_3 = yz\hat{\mathbf{x}} + zx\hat{\mathbf{y}} + xy\hat{\mathbf{z}}$ can be written both as the gradient of a scalar and as the curl of a vector. Find scalar and vector potentials for this function.

Solution:

(a)

$$\begin{aligned}
\boldsymbol{\nabla}\cdot\boldsymbol{F}_1 &= 0\\
\boldsymbol{\nabla}\cdot\boldsymbol{F}_2 &= 3\\
\boldsymbol{\nabla}\times\boldsymbol{F}_1 &= -2x\hat{\mathbf{y}}\\
\boldsymbol{\nabla}\times\boldsymbol{F}_2 &= 0
\end{aligned}$$

This means that

$$\begin{aligned} \boldsymbol{F}_1 &= \quad \boldsymbol{\nabla} \times \boldsymbol{A} \\ \boldsymbol{F}_2 &= -\boldsymbol{\nabla} V \end{aligned}$$

To find $\boldsymbol{A}$, the requirements are

$$\boldsymbol{\nabla} \times (A) = x^2 \hat{\mathbf{z}}$$
$$\downarrow$$
$$\left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right) = \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right) = 0, \quad \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right) = x^2$$

One of the vectors that satisfy this is

$$\boxed{\boldsymbol{A} = \frac{-x^2 y}{2} \hat{\mathbf{x}} + \frac{x^3}{6} \hat{\mathbf{y}}}$$

To find V, the requirements are

$$-(\boldsymbol{\nabla}(V)) = x\hat{\mathbf{x}} + y\hat{\mathbf{y}} + z\hat{\mathbf{z}}$$
$$\downarrow$$
$$\begin{cases} -\frac{\partial V}{\partial x} = x \\ -\frac{\partial V}{\partial y} = y \\ -\frac{\partial V}{\partial z} = z \end{cases}$$

One of the scalars that satisfy this is

$$\boxed{V = -\frac{1}{2}(x^2 + y^2 + z^2)}$$

(b) The divergence and curl of $\boldsymbol{F}_3$ is

$$\begin{aligned} \boldsymbol{\nabla} \cdot \boldsymbol{F}_3 &= 0 \\ \boldsymbol{\nabla} \times \boldsymbol{F}_3 &= 0 \end{aligned}$$

This means that $\boldsymbol{F}_3$ can be written both as the gradient of a scalar, and as the curl of a vector.

$$\begin{aligned} \boldsymbol{F}_3 &= -\boldsymbol{\nabla} V \\ \boldsymbol{F}_3 &= \boldsymbol{\nabla} \times \boldsymbol{A} \end{aligned}$$

The conditions to be met are then

$$-(\boldsymbol{\nabla}(V)) = yz\hat{\mathbf{x}} + xz\hat{\mathbf{y}} + xy\hat{\mathbf{z}}$$

which is satisfied by, for example

$$\boxed{V = -xyz}$$

The conditions for the vector are

$$\nabla \times (A) = yz\hat{\mathbf{x}} + xz\hat{\mathbf{y}} + xy\hat{\mathbf{z}}$$

which is satisfied by, for example

$$\boxed{\boldsymbol{A} = \frac{xz^2}{2}\hat{\mathbf{x}} + \frac{yx^2}{2}\hat{\mathbf{y}} + \frac{zy^2}{2}\hat{\mathbf{z}}}$$

□

Problem 1.51

Curl-less fields:

(a) $\nabla \times \boldsymbol{F} = 0$ everywhere.

(b) $\int_a^b \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{l}$ is independent of path, for any given end points.

(c) $\oint \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{l} = 0$ for any closed loop.

(d) $\boldsymbol{F}$ is the gradient of some scalar function: $\boldsymbol{F} = -\nabla V$

Show (d) ⇒ (a), (a) ⇒ (c), (c) ⇒ (b), (b) ⇒ (c), and (c) ⇒ (a).

Solution:

(d) ⇒ (a):

$$\nabla \times (-\nabla V) = \nabla \times \left(\frac{\partial V}{\partial x}\hat{\mathbf{x}} + \frac{\partial V}{\partial y}\hat{\mathbf{y}} + \frac{\partial V}{\partial z}\hat{\mathbf{z}}\right) =$$

$$= \left(\frac{\partial \frac{\partial V}{\partial z}}{\partial y} - \frac{\partial \frac{\partial V}{\partial y}}{\partial z}\right)\hat{\mathbf{x}} + \left(\frac{\partial \frac{\partial V}{\partial x}}{\partial z} - \frac{\partial \frac{\partial V}{\partial z}}{\partial x}\right)\hat{\mathbf{y}} + \left(\frac{\partial \frac{\partial V}{\partial y}}{\partial x} - \frac{\partial \frac{\partial V}{\partial x}}{\partial y}\right)\hat{\mathbf{z}} =$$

$$= \left(\frac{\partial^2 V}{\partial y \partial z} - \frac{\partial^2 V}{\partial z \partial y}\right)\hat{\mathbf{x}} + \left(\frac{\partial^2 V}{\partial z \partial x} - \frac{\partial^2 V}{\partial x \partial z}\right)\hat{\mathbf{y}} + \left(\frac{\partial^2 V}{\partial x \partial y} - \frac{\partial^2 V}{\partial y \partial x}\right)\hat{\mathbf{z}} = 0 \quad \square$$

(a) ⇒ (c)

Stokes' Theorem

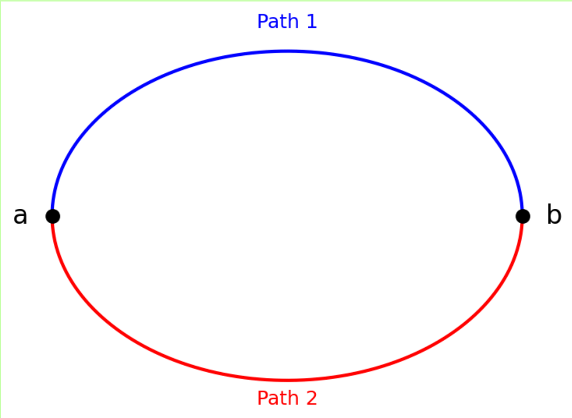
$$\int_S (\nabla \times \boldsymbol{F}) \cdot \mathrm{d}\boldsymbol{a} = \oint_P \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{l} = 0 \quad \square$$

(c) ⇒ (b)

$$\int_{a_{\text{Path 1}}}^{b} \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{l} + \int_{b_{\text{Path 2}}}^{a} \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{l} = \oint_{P} \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{l} = 0$$

$$\downarrow$$

$$\int_{a}^{b} \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{l} + \int_{b}^{a} \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{l} \rightarrow \int_{a}^{b} \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{l} - \int_{a}^{b} \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{l} \rightarrow \int_{a_{\text{Path 1}}}^{b} \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{l} = \int_{a_{\text{Path 2}}}^{b} \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{l} \quad \square$$



(b) $\Rightarrow$ (c), same as (c) $\Rightarrow$ (b), but reversed $\square$

(c) $\Rightarrow$ (a), same as (a) $\Rightarrow$ (c) (Stokes' Theorem) $\square$

$\square$

Problem 1.52

Divergence-less fields

(a) $\boldsymbol{\nabla} \cdot \boldsymbol{F} = 0$ everywhere.

(b) $\int \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{a}$ is independent of surface, for any given boundary line.

(c) $\oint \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{a} = 0$ for any closed surface.

(d) $\boldsymbol{F}$ is the curl of some vector function: $\boldsymbol{F} = \boldsymbol{\nabla} \times \boldsymbol{A}$

Show (d) $\Rightarrow$ (a), (a) $\Rightarrow$ (c), (c) $\Rightarrow$ (b), (b) $\Rightarrow$ (c), and (c) $\Rightarrow$ (a).

Solution:

(d) $\Rightarrow$ (a)

$$\boldsymbol{\nabla} \cdot (\boldsymbol{\nabla} \times \boldsymbol{A}) = \boldsymbol{\nabla} \cdot (\boldsymbol{\nabla} \times (A)) =$$

$$= \frac{\partial^2 A_z}{\partial y \partial x} - \frac{\partial^2 A_y}{\partial z \partial x} + \frac{\partial^2 A_x}{\partial z \partial y} - \frac{\partial^2 A_z}{\partial x \partial y} + \frac{\partial^2 A_y}{\partial x \partial z} - \frac{\partial^2 A_x}{\partial y \partial z} =$$

$$= \left(\frac{\partial^2 A_z}{\partial y \partial x} - \frac{\partial^2 A_z}{\partial x \partial y}\right) + \left(\frac{\partial^2 A_x}{\partial z \partial y} - \frac{\partial^2 A_x}{\partial y \partial z}\right) + \left(\frac{\partial^2 A_y}{\partial x \partial z} - \frac{\partial^2 A_y}{\partial z \partial x}\right) = 0 \quad \square$$

(a) $\Rightarrow$ (c)

Divergence Theorem

$$\int_V \boldsymbol{\nabla} \cdot \boldsymbol{F} \, \mathrm{d}\tau = \oint_S \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{a} = 0 \quad \square$$

(c) $\Rightarrow$ (b) For a boundary line that splits a closed surface into two...

(b) $\Rightarrow$ (c), same as (c) $\Rightarrow$ (b), but reversed $\square$

(c) $\Rightarrow$ (a), same as (a) $\Rightarrow$ (c) (Stokes' Theorem) $\square$

$\square$

Problem 1.53

Problem 1.54

Problem 1.55

Problem 1.56

Problem 1.57

Problem 1.58

Problem 1.59

Problem 1.60

Problem 1.61

Problem 1.62

Problem 1.63

Problem 1.64

Problem 1.65